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Household Vulnerability, Bank Solvency, and Policy Design in the Middle East War

Claudio Barbieri, Rafel Moya, and Laura Valderrama

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Household Vulnerability, Bank Solvency, and Policy Design in the Middle East War

Prepared by Claudio Barbieri, Rafel Moya, and Laura Valderrama*

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ABSTRACT: We develop a framework that links household balance sheets, bank solvency, and consumption to quantify the transmission of macroeconomic shocks and the effects of policy interventions. Using the 2023 Household Finance and Consumption Survey and supervisory bank-level data for 20 euro area countries, we project household default risk from observed arrears and introduce a cost-of-living-adjusted debt-service ratio. This measure outperforms the conventional debt-service ratio in predicting default in a statistical horse race, with delinquency risk more than doubling above a 70 percent threshold. Under a protracted Middle East war scenario, euro area households are resilient on average but highly heterogeneous: about one-third become overburdened, bank capital falls by roughly 125 basis points through higher household credit-loss provisions, and nearly 30 percent become consumers-at-risk, with a model-implied consumption-at-risk of about 5.4 percent of aggregate consumption. Counterfactual policy experiments reveal important trade-offs. *Narrowly targeted* transfers minimize household distress per euro spent, *partially targeted transfers* produce the largest gains in bank resilience, and temporary interest rate relief for highly stretched borrowers can, under some conditions, mitigate bank capital losses. These results highlight the importance of targeting in the design of policies to support household welfare and financial stability.

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Introduction and Related Literature

The outbreak of the Middle East war in February 2026 has demonstrated clearly the economic consequences of geopolitical risk.[1] While government interventions have mitigated some of the immediate effects on households, a protracted conflict could generate additional adverse shocks to households, the banking sector, and the real economy. Rising living costs and elevated lending rates are increasing pressure on household balance sheets, against the backdrop of a persistent housing affordability challenge faced by many households. Under these conditions, households may struggle to service their debt obligations, resulting in higher default rates and reduced consumption.

The transmission channels of household distress to financial stability and real activity are economically significant. First, households accounted for approximately 40 percent of all outstanding bank loans in the euro area in 2025. At around EUR 7.5 trillion, household lending represents roughly one-half of euro area GDP, making households' debt-servicing capacity a key determinant of banking sector resilience. Second, household expenditure accounts for more than one-half of euro area GDP. The Middle East conflict has already weighed on private consumption, with real consumption growth slowing to 1.3 percent year-on-year in June 2026, compared to an average 1.6 percent in 2024. A prolonged period of elevated energy prices, tighter financial conditions, and weaker confidence could further erode households' real disposable incomes, amplify financial vulnerabilities, and dampen aggregate demand (Dausa et al., 2026).

This is in the context of elevated interest rates in Europe despite a significant decline in inflationary pressures over the last two years. Lending rates for new mortgages in the euro area stood at 3.5 percent in June 2026 (0.7 percent in real terms), relative to 1.6 percent pre-Covid (0.4 percent in real terms). Outside the euro area, rates are at 4.7 percent in Czechia and 7.7 percent in Hungary —equivalent to 3.6 and 5.6 percent in real terms, respectively, compared with 0.2 and 1.2 percent pre-Covid. Interest rates that remain elevated relative to the pre-Covid period are adding to pressure on household finances and increasing their vulnerability to economic shocks. The prevalence of floating-rate mortgages varies considerably across countries, with more than half of mortgage borrowers holding adjustable-rate mortgages in two-thirds of euro area countries (Figure 1). Households with floating-rate mortgages, consumer loans and new borrowers are more vulnerable to hikes in interest rates.

House prices in Europe also remain elevated, marking their seventh consecutive quarter of year-over-year growth in the euro area as of Q1 2026. House prices grew by an average of 5 percent between 2025Q1 and 2026Q1, with five countries exceeding 10 percent average growth. While some moderation in the pace of growth was observed after the 2022 peak, prices have significantly outpaced income growth making affordability a major concern. In the euro area, the price to income (PTI) ratio stands 8 percent above the 2015 ratio, with seven countries exceeding 20 percent. This implies higher leverage for borrowers holding mortgages, which increases their vulnerability to shocks (Figure 1).

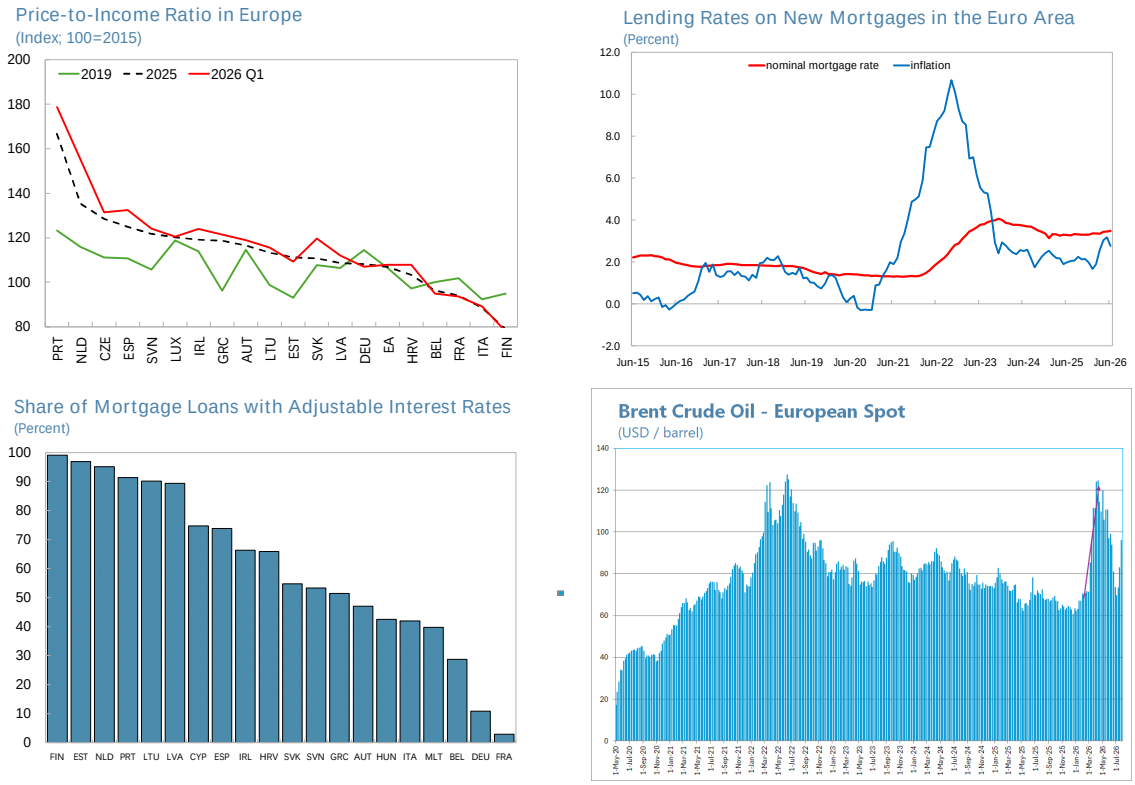
In addition to higher debt servicing costs, households have been hit by a sharp increase in energy prices. European spot Brent crude oil prices peaked at around 70 percent above their pre-conflict level in May 2026, while wholesale natural gas prices in Europe rose by 85 percent by end-July 2026. Although energy prices eased from their mid-2026 peak, at the time of writing they remain well above pre-conflict levels and continue to be highly volatile amid persistent geopolitical uncertainty. Consequently, the risk of renewed energy price pressures remains elevated. At the same time, while several major central banks have paused interest rate increases, expectations of further policy tightening by the end of 2026 persist. This could

[1] The Middle East war began on February 28, 2026, following coordinated U.S. and Israeli strikes against military and strategic targets in Iran, triggering a broader regional conflict with significant implications for commodity prices, financial markets, and economic activity.

further raise borrowing costs for households, particularly in countries with a high prevalence of adjustable-rate loans. Together, elevated energy costs and higher interest rates are likely to tighten household budget constraints, reduce real disposable income, and weigh on consumer confidence and consumption.

At the same time, European governments responded rapidly to the energy price surge triggered by the war in the Middle East. According to the European Commission's Spring 2026 forecast, budgetary measures adopted or credibly announced by early May amounted to about EUR 14.5 billion, or 0.07 percent of EU GDP in 2026, with the cost rising to EUR 38.6 billion (0.2 percent of GDP) should the measures be extended to end-2026. Bruegel's tracker points to a broadly similar order of magnitude, with more than EUR 11.8 billion committed by mid-June. The instruments deployed have consisted mainly of cuts in excise duties and VAT on fuels, electricity, and gas, complemented by sectoral fuel rebates and direct support to households. As the shock has proved more persistent than initially assumed, several countries have rolled short-term relief into multi-month packages, while some have cut fuel taxes to—or below—the floors set by the Energy Taxation Directive.

Figure 1. Price to Income Ratio and Lending Rates across Europe



Sources: Haver, OECD, 2021 HFCS; and authors' calculations.

The guiding principle is that support measures should be targeted, timely, and temporary so as to preserve price signals, avoid stimulating aggregate energy demand, and contain fiscal costs at a time when fiscal space is constrained. In practice, however, policy responses have deviated significantly from this benchmark. Around three-quarters of the support measures included in the European Commission's estimates, and more than half of those tracked by Bruegel, have taken the form of untargeted, price-distorting interventions, such as broad reductions in excise duties and indirect taxes. Also, the cost-effectiveness of targeted fiscal support remains an open question. Moreover, while policy recommendations emphasize directing support toward vulnerable households, there is little consensus on how vulnerability should be measured. Vulnerability may be associated with low income, high leverage,

limited savings, elevated debt-servicing burdens, or high marginal propensity to consume. This makes the design of well-targeted support measures particularly challenging.

The aim of the paper is fivefold. First, to quantify the ability of European households to service their debt obligations and maintain consumption under adverse macrofinancial conditions, including a protracted Middle East war scenario, using microdata. Second, to assess banking system resilience to a deterioration of the retail portfolio using supervisory data. Third, to evaluate the impact on the real economy through the consumption channel, as overburdened households cut back on consumption to meet their budget constraint. Fourth, to conduct a cost-benefit analysis of various fiscal policies aimed at shielding households from the rise in energy prices, with different targeting schemes. Finally, to simulate the impact of various debt relief policies to shield borrowers from the increase in interest rates.

Our paper uses microdata to project financial resilience and consumption patterns by exploiting the cross-sectional variation across households conditional on financial indicators of household stress across the euro area. This contrasts with most macrofinancial studies, where household debt indicators, including debt to income ratios and debt servicing ratios are calculated using information from national account statistics. However, this aggregated view masks heterogeneity about the distribution of leverage and debt burden across households. If debt is concentrated in highly leveraged households, vulnerable to shocks, and holding a significant share of bank debt, financial stability risks increase. Similarly, if households with high marginal propensity to consume are vulnerable to shocks and represent a sizable share of aggregate consumption, the likelihood of a demand-driven recession increases with deleterious effects on the real economy.

Our results suggest that, at the aggregate level, euro area households appear relatively resilient under adverse macrofinancial conditions. This aggregate resilience masks substantial heterogeneity across the income distribution and across national mortgage systems. Low-income households experience pronounced financial strain and exhibit large prospective reductions in consumption. Median-income households are, on average, more resilient but tend to carry higher leverage; when these households become financially distressed, the resulting losses have economically meaningful implications for bank capital. Using supervisory data, we find that aggregate capital depletion in the euro area banking system could reach about 125 basis points under a severe Middle East scenario, with large heterogeneity across countries. Cross-country comparisons should be interpreted with caution because contract structures, amortization patterns, tax treatment, fixed-versus-floating-rate shares, and refinancing practices differ across countries.

We also document a sizeable and tight link between household financial fragility and aggregate consumption. Our simulations identify a subset of households we term *consumers-at-risk*, defined as those for whom the sum of debt service, rents, and total consumption exceeds gross household income. In the severe Middle East scenario, consumption by these households declines sharply, implying an aggregate consumption contraction of about 5.4 percent by 2026 relative to 2025 levels. Financially stretched households are not only more likely to default, but also more likely to curtail consumption in an attempt to avoid default.

Our policy simulations yield several important policy implications. The results suggest that narrowly targeted fiscal support, directed to households in the bottom income tercile, is the most economically cost-effective intervention. Narrow policies can deliver benefits up to 5 times more than broader policies for the same fiscal cost depending on the country, with an average of twice across countries. By contrast, support measures that also cover middle-income households appear to be more effective in preserving financial stability. Partially targeted policies can deliver benefits up to 10 times more than narrow policies for the same fiscal cost depending on the country, with an average of over twice across countries. Our results show that bank characteristics and borrower heterogeneity play a crucial role in shaping the most cost

efficient policy response. The comparative performance of the simulated instruments depends on the policy objective and on country-specific distributions of income, debt, and liquidity. The exercise does not provide a complete welfare ranking and should not be interpreted as recommending a uniform policy across countries.

Related Literature

This paper contributes to several strands of the household finance and financial stability literature. It builds on the seminal work of Campbell (2006), which established household finance as a distinct field by emphasizing that households face borrowing constraints, labor income risk, housing wealth considerations, and financial decision-making challenges that are not adequately captured by representative-agent frameworks. It also relates to the comprehensive review of Gomes, Haliassos, and Ramadorai (2021), which highlights the central role of household heterogeneity in shaping portfolio allocation, borrowing, mortgage choice, and financial resilience. More broadly, the paper connects to the literature on housing, leverage, and household balance sheets, including Cocco (2005), who incorporates housing into life-cycle portfolio choice, and Mian and Sufi (2014), who demonstrate the macroeconomic importance of household indebtedness and debt overhang. It is also related to the literature on household financial vulnerability and financial stability, including Ampudia, van Vlokhoven, and Żochowski (2016), who use Household Finance and Consumption Survey (HFCS) microdata to identify financially distressed households and quantify the implications for banking sector losses.

More specifically, the paper contributes to the growing literature that uses household-level microdata to assess the resilience of households to adverse shocks (Holló and Papp, 2007; Bech et al., 2021; Bank of England, 2023; Semjonovs et al., 2022; Martinello et al., 2023). While these studies typically rely on predefined measures of financial distress, our analysis employs a statistical "horse-race" framework to identify the indicator and threshold that best discriminate between households that default and those that remain current on their debt obligations. The paper is closer to Valderrama et al. (2023), which quantifies household vulnerabilities and their implications for financial stability using survey microdata. Relative to the existing literature, we exploit arrears information contained in the 2021 HFCS to estimate household default probabilities directly from observed credit events rather than from simulated default thresholds. This approach allows us to incorporate richer information on household financial conditions, including debt-service burdens, liquidity buffers, and cost-of-living pressures, while controlling for loan, household, and borrower characteristics.

The paper also contributes to the literature linking household vulnerabilities to banking sector resilience. Previous studies combine household survey data with macroeconomic scenarios to assess the implications of household distress for banks (Gross and Población, 2017; Jurča et al., 2020; Neugebauer et al., 2021; ECB, 2022). A distinguishing feature of our framework is that it relies on observed default outcomes, rather than on assumed transition rules between distress and default.[2] This enables us to project expected credit losses across the full spectrum of loan impairment stages and to quantify the impact of household credit deterioration on bank provisions and regulatory capital in a manner consistent with the forward-looking principles of IFRS 9. To the best of our knowledge, this is the first paper to combine the 2021 HFCS with supervisory bank-level data to conduct household balance-sheet stress tests across euro area countries using observed default behavior and an IFRS 9-consistent framework.

[2] These studies assume that if liquid assets cover less than twelve months of debt and basic consumption expenditure, a household has negative disposable income and is considered to be in distress; if liquid assets cover less than one month of debt and basic consumption expenditure, such a household defaults.

The paper further relates to the literature on consumption dynamics in the presence of income, inflation, and interest-rate shocks (Druedahl and Jørgensen, 2020; Foulis et al., 2026). Unlike existing studies, however, we explicitly model the interaction between household finances, credit risk, and consumption decisions. This allows us to identify both households at risk of default and consumption at risk, defined as the amount of consumption that financially stretched households need to reduce in order to remain within their budget constraints. By integrating household credit risk and consumption responses within a unified framework, we capture an important transmission channel through which macrofinancial shocks affect both financial stability and aggregate demand.

Finally, the paper contributes to the emerging literature on the economic consequences of the Middle East war and the policy responses adopted to mitigate its effects. While recent studies and policy trackers document the fiscal measures introduced to shield households from rising energy and living costs (Bruegel, 2026; IMF, 2026a),[3] evidence on the relative effectiveness of alternative targeting schemes remains limited. Moreover, there is little analysis of financial-sector policies aimed at alleviating debt-servicing pressures through loan restructuring or temporary interest-payment relief. We address this gap by applying our framework to a protracted Middle East war scenario characterized by persistent increases in energy prices, tighter financial conditions, and weaker economic activity. We quantify the resulting effects on household default rates, bank capital, and aggregate consumption, and evaluate a range of fiscal and financial policy interventions, including income transfers under alternative targeting schemes and interest relief programs for financially stretched borrowers. By comparing their effectiveness in reducing household distress, preserving consumption, strengthening financial stability, and containing fiscal costs, the paper identifies the policy combinations that deliver the greatest benefits per unit of public resources.

Data Sources

The paper combines two rich databases. First, we use microdata from the latest (2023) HFCS to project households' financials and estimate their financial burden and consumption flows. Second, we use bank-level supervisory data from the 2023 EU-wide stress test to map the projections of households' credit risk on default rates of the retail portfolio, compute loan loss provisions and quantify bank capital impact.

Household Survey Data

The HFCS—a comprehensive ECB-led survey covering all 20 euro-area countries plus the Czech Republic and Hungary—provides insights into how household debt and assets evolved up to 2022. It is the best available data source to study household financial fragility and perform household stress test exercises in a consistent manner for euro area countries. The survey was released in 2023 (fourth wave; previous surveys were released in 2010, 2014 and 2017) and conducted with reference periods not fully harmonized across countries. In particular, the reference period for Finnish data is 2020, for most countries 2021, and for Greece 2022. After the paper was completed, a fifth 2023 HFCS wave, covering 2023, was released in June 2026.[4]

The survey contains detailed information on loan characteristics, including contract type (fixed versus floating-rate), loan-to-value (LTV) ratio, original loan amount, outstanding balance, maturity, debt-service obligations, and loan purpose, among other variables. It covers approximately 83,000 households across the euro area. Household population weights are used for household-level estimation and aggregation; individual records are used to model employment transitions within households. Household-level

[3] See 2026 European energy crisis fiscal response tracker and IMF Global Policy Tracker: How Countries are Responding to the Economic Impact of the War in the Middle East

[4] The newly released survey data contained incorrect weights for a few countries; however, this issue does not affect the vintage of the data used in this paper.

information includes demographic characteristics (such as age, gender, and number of children), savings by instrument, income by source, wealth by asset class, private and public financial assistance, indicators of credit constraints, and expenditure across consumption categories, among other variables. The household data are complemented by approximately 200,000 individual records covering all household members aged 16 and above. These records contain information on socio-demographic characteristics, educational attainment, financial literacy, employment status, labor income, type of employment contract, and pension entitlements, among other variables. We exploit the individual-level data to allocate unemployment shocks across household members and to control for household composition and individual-specific characteristics in the econometric analysis.

The 2021 HFCS data show that European households' leverage and debt service burdens varied widely in the region and by income groups, but in general remained manageable or even improved compared to earlier survey waves. European households' ability and propensity to take on debt differ markedly by country. In the euro area as a whole, about 43 percent of households carry debt (unchanged from 2017), but this ranges from roughly one-fifth of households in Greece to about two-thirds in higher-debt countries like Ireland.

Overall, median household leverage—measured as the ratio of total debt to total assets for indebted households—stood around 23 percent in the survey-based estimates (compared with 25 percent in 2017). Among those with debt, typical leverage ratios range from well below the European average in some countries to well above it in others. For example, Lithuanian households have a median debt-to-assets ratio of only 6 percent, reflecting comparatively low leverage. By contrast, in countries with historically high mortgage usage such as the Netherlands, households tend to be far more leveraged—a larger share of families carry mortgages (over 60 percent, the highest in Europe), and they borrow heavily relative to the value of their homes and incomes. In general, northern and western European economies (e.g. the Netherlands, Luxembourg) exhibit higher household leverage due to costlier housing markets and more accessible credit, whereas some southern and eastern European countries have lower average leverage because home prices and mortgage penetration are lower or borrowing standards stricter. For instance, nearly 88 percent of all household debt in the euro area is in home mortgage loans; countries where homeownership is common and credit easily available see more households taking large mortgages (raising leverage), while in places where home purchase is often debt-free or constrained by credit supply, typical leverage remains modest.

Cross-country wealth disparities also play an important role. While median household net wealth stands at about EUR 123,000 across countries, it ranges from roughly EUR 31,000 in Latvia to over EUR 700,000 in Luxembourg. In wealthier countries, households can typically sustain higher absolute debt levels and hold larger asset buffers. By contrast, in lower-wealth countries, many households carry little debt but also have fewer assets, which tempers measured leverage. When measured against income, household leverage—as captured by the debt-to-income (DTI) ratio—exceeds 70 percent in the euro area on average, peaking at around 220 percent in the Netherlands, compared with less than 30 percent in Austria, Latvia, Lithuania, and Croatia (Figure 2). The financial burden of households, measured by the debt-service-to-income ratio (DSTI), the share of gross household income devoted to debt payments – was roughly 13 percent for indebted households in the euro area. This average, however, masks significant cross-country differences, ranging from around 5 percent in Lithuania to over 20 percent in Cyprus.

Figure 2. Distribution of Household Leverage, Debt Service, and Financial Vulnerability



Sources: 2021 HFCS and authors' calculations.

There is a pronounced gap between lower-income and higher-income households in terms of leverage and debt service ratios. While higher income households are more likely to hold debt (ranging from 25 percent to 60 percent across income terciles), leverage peaks for upper-median households at over 90 percent of debt to income compared to under 50 percent for the lowest tercile. This suggests that while the very poorest tend to borrow less (often constrained by low credit access or already minimal assets), middle-class families in their prime home-buying years carry the heaviest debt loads relative to income, whereas the richest households can rely more on accumulated wealth and face lower relative debt levels.

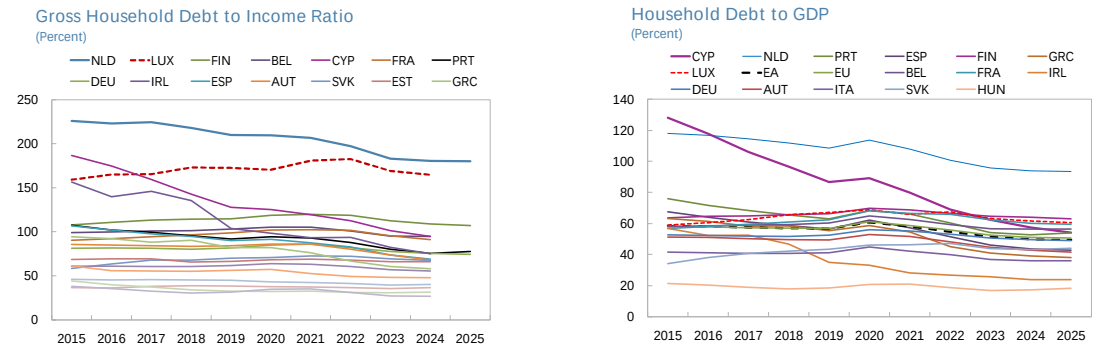
A key insight from the 2021 HFCS is the stark gap in debt affordability between lower- and higher-income households. There is a negative correlation between debt burdens and income. Low-income borrowers devote a far larger share of their income to debt repayment: in the bottom income tercile, the median debt payment eats up over 20 percent of monthly income—a heavy burden that leaves little financial buffer. By

contrast, in the top tercile of earners, the median debt payment is under 9 percent of income, reflecting both higher incomes and the ability to secure better loan terms or make larger down-payments.

Debt burden measures are indicators of household vulnerability. However, to predict the household probability of default we need all household characteristics affecting credit risk, including information on whether a household has defaulted on its debt. The 2021 HFCS survey is the first vintage that has data on late or missed payments for all countries. In particular, households were asked if they were late or missed payments with financial institutions over the previous 12 months, and if so, whether they were overdue payments by more than 90 days. We use this information to predict credit default rates. The 2021 HFCS data show that for about 60 percent of households regular expenses exceed their income level, with lower levels in Germany (one third of households) than in Croatia (four out of five households). These figures reflect household circumstances as of 2021; in the analysis, household incomes and expenses are updated to 2025 levels. Households with adjustable-rate mortgages are more vulnerable to interest shocks as their mortgage reprice to benchmark rates. In the survey-based estimates, about 60 percent of mortgage-holding households have floating-rate mortgages with the share being the highest in Finland, Lithuania, Netherlands and Portugal (at over 90 percent of mortgages) and lowest in France (3 percent of mortgages) (Figure 2). These survey-based shares should not be interpreted as the current composition of outstanding mortgages. In Portugal, recent lending has shifted toward mixed-rate contracts, which typically have an initial fixed-rate period of two to five years before reverting to a floating rate, and these contracts account for around half of outstanding mortgages. Mixed-rate contracts limit immediate repricing risk during the fixed period but remain exposed thereafter. Survey-based projections that do not capture this shift may therefore overstate near-term interest-rate sensitivity in Portugal.

While the survey provides a rich and detailed picture of heterogeneity in household finances, it is subject to a production lag of three to five years. Aggregate data from national statistics indicate that, in many countries, households have deleveraged since the 2022 peak, with the gross household country-level debt-to-income ratio declining from 84.0 to 75.6 percent and household debt-to-GDP falling from 55.8 to 50.7 percent (Figure 3). More broadly, euro area private non-financial sector debt is currently close to historical lows. While we use the deleveraging information to update household balance sheets, this low level of aggregate leverage is itself an important mitigating factor for financial stability and helps explain why the median household withstands the shock even as pockets of vulnerability remain. However, aggregate data mask vulnerabilities among highly leveraged households and do not capture the sharp increase in debt-servicing costs following the onset of the monetary tightening cycle in Europe in 2022. To obtain a timely measure of household financial conditions as of 2025, we combine the 2021 HFCS microdata with aggregate information from national statistical offices, the OECD, and Eurostat covering the period 2022-2025. Following the approach of Krimmel et al. (2013), Ampudia et al. (2016), and Valderrama et al. (2023), we project household balance sheets, income, debt-service obligations, and consumption expenditures to end-2025. The resulting dataset provides the starting point for the three-year household stress testing exercise. This updating is a microsimulation, not a new observation of each household in 2025. Aggregate changes cannot fully capture shifts in the joint distribution of income, debt, assets and consumption, new borrowers, refinancing, or changes in mortgage contract types. The robustness checks in Appendix I assess estimation and prediction; they do not validate the distributional accuracy of the end-2025 updating. These limitations are particularly relevant for cross-country comparisons and households close to vulnerability thresholds.

Figure 3. Aggregate Household Debt by Country



Sources: Eurostat and authors' calculations.

Regarding household consumption, while HFCS has granular data on amount spent on consumption of goods and services by category, it does not have information on durable consumption. We estimate durable consumption by mapping the sub-components included in the HFCS survey to the consumption basket in the harmonized index of consumer prices (HICP). Using the composition of the HICP by country, we impute a level of consumption of excluded items at the household level by matching the sub-categories covered in both the HFCS and the HICP and scaling the amount spent on non-essential consumption (excluding rents) at the household level. This procedure allows taking into account key characteristics of households, such as their income level or the share of income spent on essential goods (food, utilities, rents). The imputed component is not directly observed household expenditure. Applying national consumption-basket shares cannot fully capture household-specific durable purchases or substitution across categories; consumption-at-risk estimates are consequently sensitive to this imputation.

Bank Supervisory Data

The default risk of households is matched up with the granular supervisory banking sector data of household loans at the bank level across euro area countries. The paper uses 2023 EU-wide stress test data for household exposures, IFRS 9 staging, provisions, coverage ratios, and risk-weighted assets (RWAs). The supervisory data are bank-specific, but the HFCS does not identify the bank that originated each household loan. The share of loans to total assets differed markedly across countries. Focusing on the significant institutions of euro area countries supervised by the ECB, the share of household loans was highest in Netherlands, Finland, and Estonia at over one third of bank assets, and lowest in Germany and Greece at over 10 percent of total bank assets. In terms of GDP, household loans were highest in Finland, France, Netherlands, and Spain, exceeding 160 percent of GDP. Regarding asset quality, most household exposures were classified as IFRS 9 Stage 1 loans (89 percent on average), while Stage 2 and Stage 3 exposures accounted for 8 percent and 3 percent, respectively, with significant cross-country heterogeneity.[5] The 2023 supervisory calibration is distinct from the 2021 HFCS wave label and the end-2025 household starting point. Subsequent changes in bank exposures, staging, provisioning coverage and RWAs are not fully captured by retaining this calibration.

Scenario-based exposures and provisions are calculated by distributing total exposures based on original proportions and scenario-specific shifts. The difference between total scenario provisions and the

[5] Under the IFRS 9 impairment framework, banks classify credit exposures into three stages according to their expected credit losses (ECLs) and underlying credit risk. Stage 1 includes performing exposures, Stage 2 captures exposures with a significant increase in credit risk, and Stage 3 comprises credit-impaired (non-performing) exposures

provisions reported by banks in 2023 (the “provision delta”) represents the additional capital that banks would need to set aside to cover the loan losses that would materialize in those scenarios. It is worth noting that while 2021 HFCS data covers additional countries within the European Union, the banking data refers to euro area banks.

Identifying Vulnerable Households

Adverse macrofinancial developments, including a protracted 2026 Middle East war, can affect financial stability and economic activity through two key household-related channels:

1. Banking channel: Higher energy prices, inflation, and borrowing costs increase debt servicing burdens, raising the risk of loan defaults and bank losses. This may erode bank capital and lead banks to tighten lending, amplifying the economic slowdown.

2. Consumption channel: The surge in energy and living costs erodes real household incomes, affecting particularly low-income or highly indebted households with thin buffers. As a result, they may cut discretionary spending weakening aggregate demand and reinforcing economic and financial stress.

Households may become overburdened under severe macrofinancial stress. Even in milder scenarios, rising living costs and interest rate shocks can strain household finances—especially among lower-income groups, who often struggle to adjust spending and highly leveraged borrowers. While there are numerous papers that perform simulation exercises to assess the degree to which households are vulnerable to adverse financial conditions, a common definition of financial fragility is lacking.

Some studies exploiting household-level microdata focus on the *financial margin*, defined as disposable income net of basic living expenses and loan installment payments. Households with negative financial margins are typically considered at high risk of default (Holló and Papp, 2007; Martinello et al., 2023). However, this metric has limitations. Financially distressed households may temporarily avoid default by borrowing or drawing on liquidity buffers, while others may fall into arrears despite maintaining a positive financial margin.

Other studies use the *DSTI* defined as the ratio of monthly loan installments to total disposable income to capture household financial stress. In this paper, DSTI is measured relative to gross household income, consistently with the HFCS income variable used in the analysis. Some studies use this ratio to quantify the sensitivity of households’ cash-flows to interest rate shocks (Bech et al, 2021), while others use this ratio as a proxy of household overburden rate. For instance, a DSTI ratio of 40 percent is a common borrower-based measure (BBM) used by central banks and regulators to maintain financial stability.[6] Often it also serves as a reference limit on the share of a borrower’s monthly disposable income that can be used for debt repayments to ensure prudent underwriting standards. A variation of the DSTI is the *housing cost overburden rate*, as defined by the OECD (2021) and Eurostat (2025), which classifies a household as overburdened when it spends more than 40 percent of its disposable income on housing costs. Housing costs typically include mortgage interest and principal repayments, rent, utilities, insurance, and maintenance expenses but exclude other essential payments such as non-mortgage debt repayments or food consumption. Accordingly, the 40 percent gross-income threshold used here is an analytical benchmark, not a direct application of a regulatory limit defined on net or disposable income. Regulatory DSTI definitions and thresholds vary across jurisdictions.

[6] As of December 2025, among the 17 EU countries that applied BBM DSTI limits, eight countries set the limit at 40 percent, five countries at 50 percent, two countries at 60 percent and one country each at 35 percent and 80 percent.

A related challenge is to identify which DSTI ratio best predicts default. DSTI ratios around the 30–50 percent range for mortgages have repeatedly emerged as critical thresholds in household-level data, above which loan default risk rises sharply. For instance, Michelangeli and Pietrunti, (2014) find a 30 percent threshold of distress for mortgage delinquencies in Italy. Nier et al (2019) find that mortgage default probabilities stayed low at DSTI levels up to 50 percent but then surged when DSTI exceeded 50 percent of income in Romania. Dirma, M. and Karmelavičius, J. (2025) find an inflection point between 25 and 40 percent where the marginal impact of changes in DSTI on mortgage probability of default in Lithuania is the highest. These studies use loan-level data and focus on one country, as default information available in credit registries is confidential.

Despite its widespread use, the DSTI ratio has three important limitations. First, it does not adequately capture affordability pressures faced by low-income households. Such households are more vulnerable to adverse income shocks because they have less scope to absorb declines in income without cutting essential consumption or falling below the poverty line. Yet, when vulnerability is measured solely through debt-service obligations, they may appear relatively resilient because they are less likely to hold significant bank debt. Second, the DSTI ratio ignores essential non-debt expenditures. As a result, it fails to capture the disproportionate burden that increases in the cost of necessities, such as food, energy, and housing-related expenses, impose on leveraged households, for whom debt repayments and essential consumption account for a large share of total spending. Third, the DSTI ratio is ill-suited to assessing household stress under cost-of-living shocks, where prices of essential goods rise faster than incomes, as observed during the inflation surge that followed the onset of monetary tightening in Europe in 2022.

A further challenge is defining the appropriate scope of household indebtedness when assessing vulnerability. Some cross-country studies rely on securitized mortgage data to predict default. For example, Bandoni et al. (2025) use securitized mortgage loans from the European Data Warehouse covering Spain, Ireland, Italy, and Portugal. However, the representativeness of securitized mortgages is uncertain. Moreover, such datasets exclude other forms of household borrowing, including consumer loans, credit card debt, and overdraft facilities, which typically exhibit substantially higher default rates than mortgages. We therefore argue that household vulnerability should be assessed using a comprehensive measure of indebtedness that encompasses all major forms of borrowing and is consolidated at the household level. Such an approach accounts for the joint financial position of household members and captures the potential for cross-subsidization of debt-service obligations within the household.
Studies also differ in the definition of default. Recent cross-country studies construct their own measure of default. For instance, Ampudia et al (2016), Dieckelmann and Metzler (2022), and Gross et al (2022) use euro area survey data and proxy default by the number of months that overburdened households will exhaust their savings to cover debt payments and basic expenditures. However, not all vulnerable households will default, and conversely, not all non-vulnerable households will service their debt repayments on time.

To capture household vulnerability, we use a *cost-of-living adjusted debt servicing ratio*, i.e., the debt-service and essential consumption to income ratio (DSECTI), a granular indicator of debt affordability and cost-of-living pressures. The numerator includes debt service obligations and essential consumption—namely food, energy costs, and rent—while the denominator comprises gross household income, including income from financial assets. A similar ratio is used by Bank of England (2023) to measure a cost-of-living adjusted mortgage ratio. By contrast to Bank of England (2022) our measure of essential spending is narrower on consumption (it includes food, utilities, and rents, but excludes housing maintenance, taxes, motor fuels, vehicle maintenance and communications) and to Martinello et al. (2023) where the monthly expected costs include modest standard of living variable costs defined by the Danish Financial Supervisory Authority. We measure default using missed debt payments reported at the household level. Throughout the empirical exercise, income in DSTI, DSECTI and DSTCTI denotes gross household income before personal income taxes and social contributions, not disposable income. Because debt service and consumption are financed from after-tax resources, gross-income ratios can understate cash-flow pressure.

A 70 percent DSECTI threshold therefore need not represent the same degree of strain across countries with different tax-benefit systems. The threshold is an empirical classification within this dataset, not a universal affordability limit; net-income-based comparisons would require a consistent tax-benefit adjustment.

This ratio captures key dimensions of household vulnerability to shocks, including:

- Commodity price shocks: Increases in energy and food prices, rents, and other essential expenditures that raise household living costs and reduce disposable income available for debt servicing.
- Interest rate shocks: Higher debt-servicing burdens resulting from policy rate increases, widening credit spreads, or tighter financing conditions, which affect households through the repricing of variable-rate loans, the refinancing of maturing debt, and the origination of new borrowing.
- Labor market deterioration: Reductions in household income arising from higher unemployment, lower wage growth, or broader recessionary conditions that weaken households' debt repayment capacity.

Identifying vulnerable households and understanding the pass-through of shock transmission require detailed data on debt repayment structures and consumption patterns. It is essential to segment liabilities by loan type and repricing schedule and to distinguish between essential (inelastic) and non-essential (discretionary) spending. Equations (1) and (2) illustrate how such shocks affect households' ability to meet debt obligations and maintain essential consumption. The equations show that the impact of macrofinancial shocks on household finances is highly non-linear.

$$DSTI^{h}_{T,j} = \frac{\sum_{k=1}^{N}\left(P^{h}_{t,k} + O^{h}_{t,k}\cdot i^{h}_{t,k}\right) + \sum_{s=1}^{M}\left(O^{h}_{t,s}\cdot \Delta i_{T-t,j}\,\middle|\, s = \ floating\right)}{I^{h}_{t}\cdot\left(1+\Delta I^{h}_{T-t,j}\right)} \quad \text{.....(1)}$$

$$DSECTI^{h}_{T,j} = DSTI^{h}_{T,j} + \frac{food^{h}_{t}\cdot\left(1+\Delta food^{h}_{T-t,j}\right) + utilities^{h}_{t}\cdot\left(1+\Delta energy^{h}_{T-t,j}\right) + rent^{h}_{t}\cdot\left(1+\Delta \inf_{T-t,j}\right)}{I^{h}_{t}\cdot\left(1+\Delta I^{h}_{T-t,j}\right)} \quad \text{..(2)}$$

where h denotes the household; P is the principal repayment on outstanding debt; O is the stock of outstanding debt; i is the lending interest rate; I is total gross household income; *food* denotes expenditure on food consumed at home and one-half of expenditure on food consumed outside the home; *utilities* captures spending on electricity, gas, and water; *rent* is rental expenditure; N is the total number of loans; and M is the number of loans with floating interest rates. Rental expenditures are assumed to be indexed to inflation.

The analysis is based on several conservative assumptions. First, food and utility expenditures are assumed to evolve in line with food and energy price inflation projections from the IMF Global Assumptions (GAS). Second, real consumption is assumed to grow at the rate projected in the World Economic Outlook (WEO). Third, we do not model behavioral adjustments, such as reductions in energy consumption, substitution across consumption categories, or changes in dietary patterns in response to higher prices. This assumption is relevant in the context of the recent Middle East conflict, where policy responses in many countries have largely relied on broad-based, price-distorting measures. In the no-policy stress simulations, the global energy-price shock is assumed to pass through fully to domestic household energy prices. This is a deliberately strong benchmark rather than an estimate of current country-specific retail pass-through. Regulated tariffs, taxes, subsidies, hedging, contract duration and the electricity generation mix can attenuate or delay transmission. The food-price calibration based on 2022 is also an assumption,

not evidence that the current shock propagates identically. The policy section relaxes the effective pass-through to households' net energy burden through compensating transfers; it does not estimate a new wholesale-to-retail price coefficient.

Estimating Household Default Risk

Methodology

We estimate the relationship between household vulnerability and the default risk using household-level logistic regressions. Specifically, we assess the ability of alternative stress indicators to predict credit events consistent with IFRS 9 definitions. The indicators considered include the DSTI ratio and the cost-of-living-adjusted DSECTI. For each indicator, we identify the threshold that best predicts default by conducting a horse race across alternative cut-off values and selecting the specification with the strongest econometric performance.

We use arrears as a proxy for default. To estimate the probability of falling into arrears, the logistic regressions control for income-tercile and country fixed effects, as well as a broad set of household, individual, and loan characteristics. To balance the reduction in omitted-variable bias against the risk of overfitting, we estimate a sequence of increasingly rich model specifications. This approach allows us to assess the robustness of the estimated thresholds and the stability of the coefficients of interest across alternative control sets:

- Model 1: baseline specification, includes country and household income-tercile fixed effects.
- Model 2: adds the household saving ratio.
- Model 3: adds demographic and financial variables (age, gender, education, financial knowledge, employment status, work experience, number of unemployed household members, number of dependent children).
- Model 4: adds loan-to-value (LTV) of main residence, family/public assistance, and leasing obligations.
- Model 5: adds adjustable rate, liquid assets, credit constraints, inheritance, and the net wealth-to-income ratio.

Formally, equation (3) shows the logistic model used to predict arrears. The linear index is transformed by the logistic cumulative distribution function, ensuring that predicted probabilities lie between zero and one:

$$Arrears_i = \alpha + \beta \cdot DSECTI\ overburden_i + Country_i + Income\ tercile_i + \gamma \cdot X_i + \varepsilon \quad \text{.....(3)}$$

where

- *Arrears* is a binary variable which takes the value of 1 if the household has defaulted in the previous 12 months and 0 otherwise
- *DSECTI overburden* is a binary variable equal to 1 if the DSECTI ratio exceeds a specific threshold and 0 otherwise
- *Country* and *income tercile* are fixed effects
- *X* are household and individual level controls

All model specifications include country and income-tercile fixed effects. Standard errors are clustered at the household level.

We also estimate a set of pooled logistic regressions to predict the probability of arrears exceeding 90 days, controlling for country and income-tercile fixed effects. Given the limited number of arrears observations in many countries, borrower-level controls are excluded to preserve statistical power and

avoid overparameterization. This parsimonious specification, however, limits the ability to control for cross-sectional heterogeneity in household risk characteristics. Standard errors are clustered at the country level to account for within-country correlation in the error terms.

A set of robustness checks reported in Appendix I complements our choice of measure and methodology. We benchmark DSECTI against the conventional DSTI and DTI ratios using the same specification, sample, and cross-validation folds, comparing both in-sample effects and out-of-sample discriminatory power, and validate the 70 percent threshold out of sample. We further re-estimate our baseline models on a common sample to rule out sample-composition effects, assess the sensitivity of our results to survey weighting, and check the stability of our main coefficient using a rare-events bias-corrected logit and a cross-validated regularized logit. These checks confirm that our main findings are robust to the choice of vulnerability indicator, sample, weighting scheme, and estimation method.

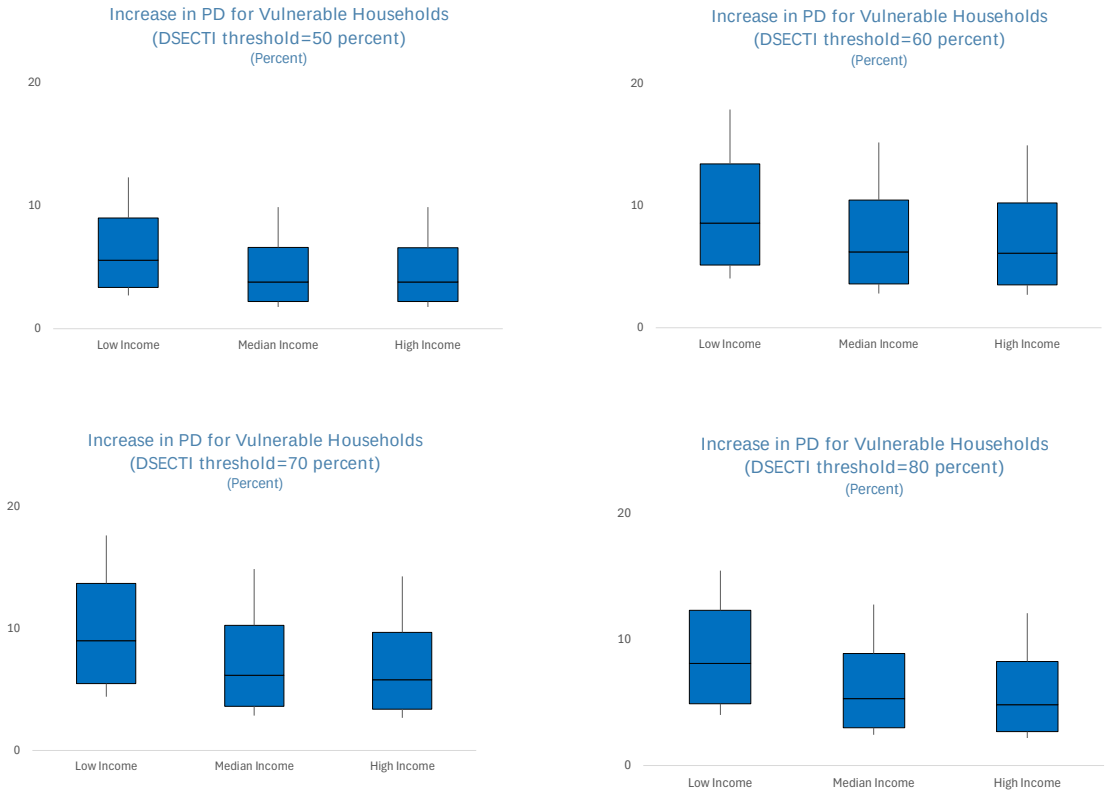
Findings

This section presents the results of the logistic frameworks for the models discussed above for the cost-of-living adjusted DSECTI ratio. This indicator turns out to be the most effective predictor of household distress. Table 1 shows the log-odds ratios of the regressions where a vulnerable household is defined as a household with DSECTI ratio exceeding 70 percent of income. These ratios can be interpreted as the usual beta, so a positive value increases the likelihood of arrears, while a negative value decreases the likelihood of arrears. The results are broadly comparable across model specifications, although the estimation sample declines as additional controls are introduced. Appendix I, Table A3, therefore re-estimates all models on a common sample finding robust results. A DSECTI indicator equal to 1 is associated with a substantially higher probability of default. Default risk declines significantly across higher income terciles, relative to the bottom income tercile. Loan impairment probabilities increase for unemployed borrowers and for borrowers relying on family or public assistance. Borrowers with a mortgage loan are more likely to default as LTV ratios increase reflecting higher leverage and more limited refinancing capacity in the presence of adverse shocks. By contrast, the savings ratio does not emerge as an economically significant predictor of default, a finding that differs from assumptions adopted in previous studies.

Table 2 shows the predicted probabilities which are defined as the likelihood of arrears evaluated at particular levels of the DSECTI ratio and income terciles, assuming that other regressors are at their means. All models suggest that households overburdened by rising debt service and basic expenditures are significantly more likely to fall into arrears on their outstanding loans. Country-level logit regressions using model 4 indicate that, in aggregate, the default rate more than doubles when households become overburdened. Specifically, the median probability of arrears rises from 5 percent to 10 percent. Also, the probability of default decreases with the income tercile of the household, with low-income households showing an 8.2 percent default rate almost twice the upper tercile at 4.8 percent. Model 4 is used for the main simulations because it adds housing leverage and transfer variables while retaining broader coverage than Model 5. Its results are also checked against specifications estimated on the broader sample and on a common estimation sample (Appendix I). The results are robust across specifications, indicating that the main findings are not sensitive to the choice of controls.

The probability of arrears varies markedly across countries. In most cases, arrears rates remain in the single digits. However, in several Southern and Eastern European countries—Cyprus, Greece, Croatia, Hungary, and Lithuania—arrears among overburdened households exceed 20 percent. These estimates may be biased upwards, as the HFCS survey was conducted during the Covid-19 crisis, when extensive forbearance measures were in place to shield households from financial distress, and therefore households could restructure loans without affecting bank capital. Across all countries, low-income households are consistently more prone to default on housing loans, particularly when they face elevated debt service burdens and rising living costs.

Figure 4. Increase in Default Rates for Vulnerable Households at Different Thresholds

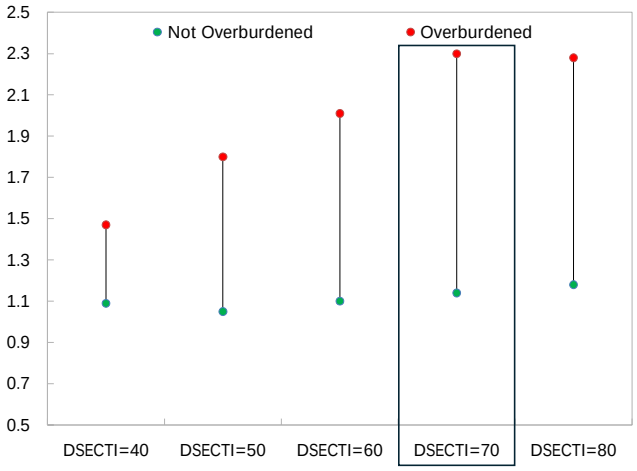


Sources: 2021 HFCS, and authors' calculations. Note: The boxplots show the cross-country distribution of the increase in the probability of arrears for overburdened households (defined as households with a DSECTI exceeding the specified threshold) relative to non-overburdened households, by income tercile (low-income, median-income, high-income) across a range of DSECTI thresholds. Probabilities are estimated from pooled household-level logit regressions controlling for borrower characteristics and country fixed effects.

We re-run the regressions across alternative DSECTI thresholds to determine which cutoff best predicts household default and is therefore most suitable for monitoring financial distress. Specifically, we assess the discriminatory power of each threshold by measuring the extent to which it maximizes the difference in the probability of falling into arrears between households above and below the threshold. Regressions are estimated using model 4 for a range of thresholds, and marginal effects are computed at both the income-tercile and country levels. Figure 4 shows the increment of the default rate between an overburdened and a non-overburdened household for DSECTI thresholds defined between 50 and 80 percent. The box plots show the distribution across countries and income terciles. The inflection point is between 60 and 70 percent depending on the country and income tercile. For instance, for the median country, and the bottom income tercile, the probability of default increases by 9 percentage points using a threshold of 70 percent relative to 5 percentage points using a threshold of 50 percent.

Similarly, cross-country logit regressions for defaults defined as arrears exceeding 90 days show that the probability of default doubles for overburdened households. Using a DSECTI ratio of 70 percent, the PD increases from 1.1 percent to 2.3 percent when households are financially strained by debt service payments and essential living expenses. Figure 5 shows the increase in probability of falling into arrears over 90 days across different DSECTI ratios. The largest difference in the probability of default is at the cut-off rate of 70 percent where the gap between overburdened and not overburdened households is 1.2 percentage points relative to 0.4 percentage points using a 40 percent threshold.

Figure 5. Probability of Arrears (>90 Days) Among Vulnerable Households
(Percent)



Sources: 2021 HFCS, and authors' calculations. Note: The chart shows the probability of arrears for overburdened households (defined as households with a DSECTI exceeding the specified threshold) relative to non-overburdened households, across a range of DSECTI thresholds. Probabilities are estimated from pooled cross-sectional logit regressions controlling for income tercile and country fixed effects.

Stress Testing Households

We apply a simulation-based framework to project household balance sheets, debt service obligations, income, and consumption under alternative macrofinancial scenarios. Household financial positions are first updated from the survey reference year (approximately 2021 for most countries) to the starting point of the stress test in 2025, using realized country-level economic outturns from national accounts and related data sources. We then project household financials over the 3-year stress testing horizon (2026-2028) under a range of macrofinancial scenarios. The quantitative results are illustrative, conditional scenario estimates, not forecasts. The emphasis on 2026 reflects the first-year shock and policy experiments; the framework also projects 2027-2028, as shown in the bank-capital results. Later-year outcomes depend on the assumed macroeconomic recovery and should not be read as evidence that vulnerabilities necessarily dissipate.

Description of the Scenarios

We consider four sets of scenarios: a baseline scenario, a protracted Middle East war scenario, a severe macro-financial stress scenario, and a series of sensitivity analyses based on combinations of macroeconomic shocks. The baseline scenario is based on the October 2025 WEO projections for the global economy and euro area countries. These projections are applied to the updated household balance sheet and income positions at the starting point of the stress test in 2025. The baseline scenario envisages a gradual recovery in the euro area, with real GDP growth projected at 1.1 percent in 2026 and 1.4 percent in 2027 following several years of subdued expansion. The recovery is supported by a progressive strengthening of investment and trade, aided by monetary policy easing and continued resilience in the services sector. Inflation is projected to continue declining toward the ECB's medium-term target, averaging between 1.9 and 2.1 percent over 2026-2028. Unemployment is expected to decline modestly, from 6.4 percent in 2025 to 6.2 percent by 2027-2028.

To assess the implications of the current geopolitical environment, we consider a protracted Middle East war scenario based on the April 2026 WEO. Relative to the January 2026 WEO Update baseline, the

scenario assumes a sustained doubling of oil and gas prices in 2026, together with an increase in food commodity prices calibrated using the oil-to-food price pass-through observed during the 2022 energy crisis. The resulting surge in energy and food costs generates broad-based inflationary pressures, leading one-year-ahead inflation expectations to increase by up to 100 basis points. Financial conditions also tighten as heightened uncertainty triggers a marked risk-off episode, raising corporate credit spreads in advanced economies by an additional 100 basis points in 2026. The larger increase in DSECTI-based vulnerability than under the FSAP geopolitical scenario reflects the assumed size and persistence of energy and food cost increases, full energy-price pass-through in the no-policy benchmark, and limited short-run compression of essential spending. These assumptions can move many households across the 70 percent threshold even when debt-service-only indicators change less. Country-specific retail price transmission may differ materially from this benchmark. The larger increase in DSECTI-based vulnerability than under the FSAP geopolitical scenario reflects the assumed size and persistence of energy and food cost increases, full energy-price pass-through in the no-policy benchmark, and limited short-run compression of essential spending. These assumptions can move many households across the 70 percent threshold even when debt-service-only indicators change less. Country-specific retail price transmission may differ materially from this benchmark.

The severe macro-financial stress scenario is based on the geopolitical scenario developed for the 2025 IMF Euro Area Financial Sector Assessment Program (FSAP) (Arnould et al., 2025). The FSAP geopolitical scenario assumes an escalation of geopolitical tensions relative to baseline, generating heightened commodity price volatility, supply-chain disruptions, adverse trade and tariff shocks, and deteriorating financial conditions. These shocks substantially weaken economic activity in the euro area, with real GDP contracting by 3 percent in 2026, approximately four percentage points below the baseline. Long-term interest rates rise sharply, with ten-year sovereign bond yields peaking at 6.5 percent in 2027, compared with around 3 percent in 2025. Labor market conditions also deteriorate significantly, with unemployment increasing by up to 4.6 percentage points above baseline levels by 2028. Consistent with the FSAP calibration, lending rates for households are derived by applying the cumulative shocks to short- and long-term interest rates over the stress-test horizon and selecting the maximum increase relative to 2021 borrowing costs.

Given the uncertainty surrounding future macro-financial developments, we further assess household resilience through a set of sensitivity analyses. These include: (i) a *tightening* scenario featuring an additional 200 basis point increase in lending rates relative to the baseline; (ii) an *income* scenario combining a 10 percent decline in gross nominal household disposable income with a 5 percentage point increase in unemployment; (iii) a *cost-of-living* scenario combining the interest-rate shock with a 20 percent increase in food and energy prices relative to the baseline; and (iv) a *combined* scenario incorporating both the interest-rate and income shocks. The magnitude of these shocks is broadly consistent with previous household stress-testing studies (Ampudia et al., 2016; Martinello et al., 2023) and the 2025 EU-wide stress-testing exercise. The average country shocks across scenarios during the first year of stress, i.e., 2026, are reported in Table 3, which sets out the seven scenarios considered (baseline, the two main scenarios, and the four sensitivity analyses) and is now presented in the main text to clarify the scenario design upfront. For ease of exposition, the household- and debt-at-risk charts (Figures 6–9) and the consumption-at-risk chart (Figure 15) display the baseline together with the two main scenarios (FSAP geopolitical and Middle East war), whereas the bank-solvency results (Figures 12–13) report all seven scenarios. Results for the four sensitivity analyses on households, debt at risk, and consumption are qualitatively similar and are available upon request.

For each scenario, we project a set of key variables that characterize the main transmission channels through which adverse macro-financial conditions affect household financial distress. These variables include labor income, employment status, debt-service obligations, interest payments, essential consumption expenditures, and household balance-sheet positions. Inflationary shocks erode real

disposable income, while higher interest rates increase debt-servicing costs for variable-rate loans, refinanced mortgages, and newly originated credit. Unemployment shocks operate through labor income losses and transitions to unemployment benefits. Cost-of-living shocks are captured through higher expenditures on essential goods, particularly food and energy, which may increase more rapidly than wages. By jointly projecting these components, the framework captures the evolution of household repayment capacity and identifies borrowers that become financially vulnerable under adverse macro-financial conditions.

Unemployment shocks are allocated at the individual level within the household using a Monte Carlo simulation framework that accounts for the employment status of the borrower and contract type. Specifically, employed borrowers are exposed to random unemployment shocks calibrated to match the projected aggregate unemployment path under each scenario. Consistent with empirical evidence, fixed-term contract holders are eleven times more likely to face unemployment than those with permanent contracts.[7] While cross-country differences in labor market segmentation and wage transitions between contract types might impact the migration rates to unemployment, this assumption provides a tractable approximation. When we draw an employed individual to unemployment, we estimate the resulting disposable income depending on unemployment benefits. We assume that unemployment benefits are at about 10 percent of initial income. We conduct robustness checks for country-based eligibility for unemployment benefits, but results are quantitatively similar.

Throughout the combined projection period, the country-specific survey reference dates through 2028, some outstanding loans mature. We replace maturing loans by new issuances to keep banks' household loan portfolio constant. Loan dynamics are modelled such that maturing loans are replaced with new ones maintaining the same DTI ratio at origination and issued at prevailing market rates. Floating-rate mortgages are reset according to contractual terms and repriced at prevailing market rates.

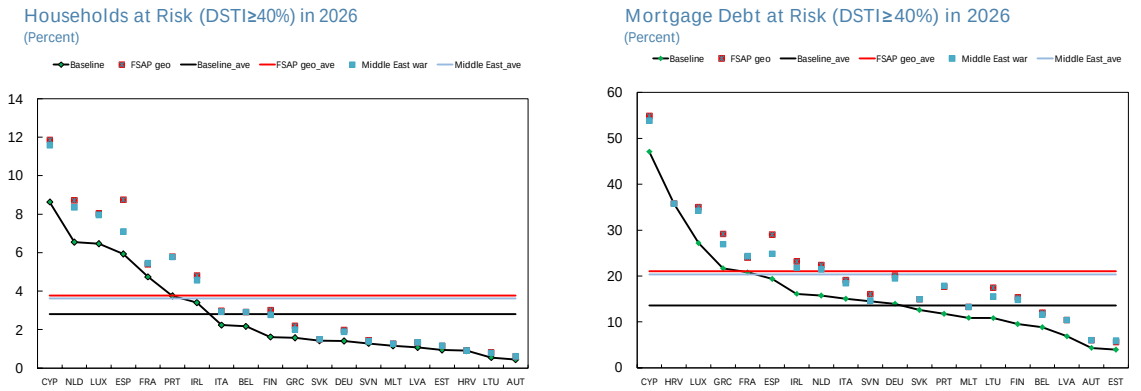
Findings

Households at Risk

Using the standard debt-service-to-income ratio (DSTI), the simulations suggest that, by 2026, the share of households at risk (DSTI ≥ 40%) could rise moderately to 4 percent, though with wide cross-country variation (Figure 6). Driven by higher interest rates and falling incomes, several high-debt euro area economies could see their at-risk household share increase by over one third under the FSAP geopolitical scenario. For example, in Cyprus the fraction of households with DSTI ≥40% jumps from about 8.6 percent under the baseline to 12 percent. Netherlands shows a similar surge (from 6.5 percent to 9 percent). In contrast, lower-leverage markets such as Germany experience only modest increases (remaining around 2 percent of households at risk in the worst case), while Austria and Lithuania stay below 1 percent even under the toughest scenario. These outcomes reflect differences in household debt levels, mortgage structures (e.g., prevalence of variable rates), housing ownership rates (e.g., low in Germany) and initial financial buffers across countries.

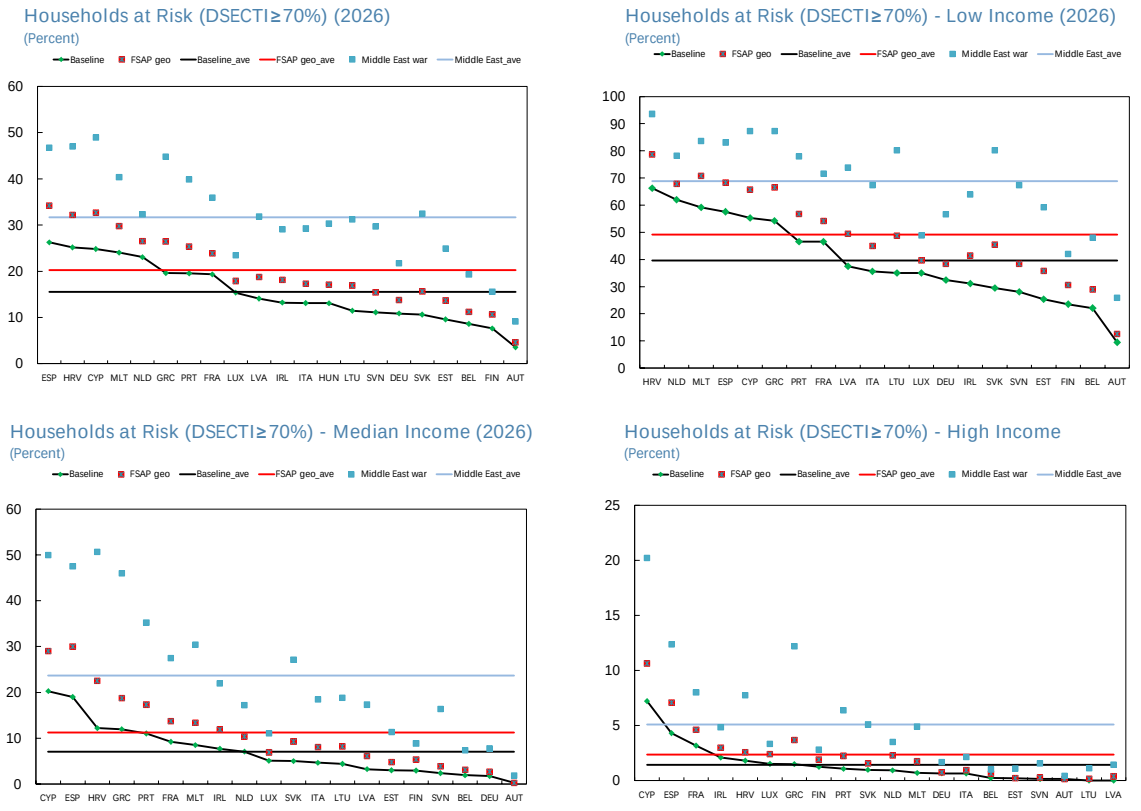
[7] This estimate is based on the share of transitions from employment to unemployment observed between 2021 and 2024, according to the Labor Force Survey of a major euro area country.

Figure 6. Simulation Results—Standard Measure of Financial Stress



Sources: 2021 HFCS survey and authors' calculations. Note: The black line shows country-level aggregates of household projections for 2026 under the Baseline scenario, based on the October 2025 WEO baseline forecast. The dots represent alternative scenarios: (i) "FSAP geo", corresponding to the geopolitical scenario developed for the 2025 Euro Area FSAP; and (ii) "Middle East War", aligned with the April 2026 WEO severe scenario, which assumes a larger and more persistent commodity price shock. Horizontal lines indicate the cross-country average under each scenario.

Figure 7. Households at Risk by Income Level—Cost-Adjusted Measure of Financial Stress

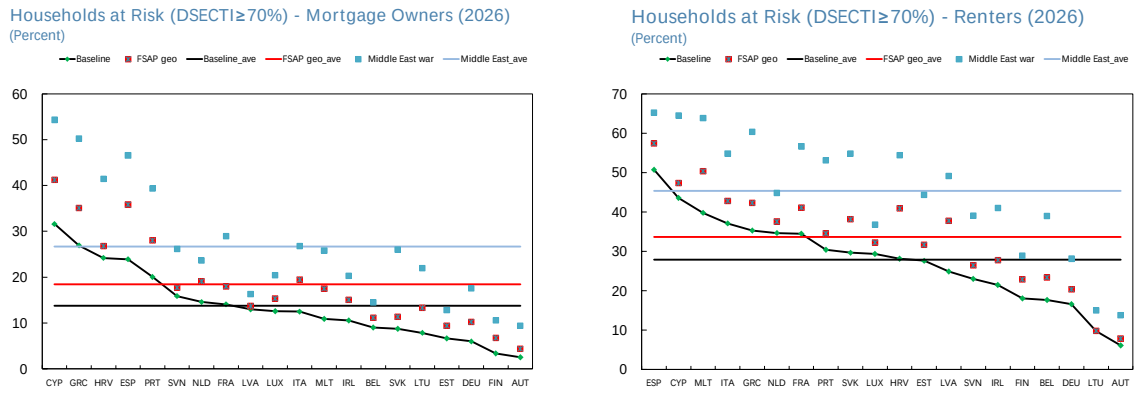


Sources: 2021 HFCS survey and authors' calculations. Note: The black line shows country-level aggregates of household projections for 2026 under the Baseline scenario, based on the October 2025 WEO baseline forecast. The dots represent alternative scenarios: (i) "FSAP geo", corresponding to the geopolitical scenario developed for the 2025 Euro Area FSAP; and (ii) "Middle East War", aligned with the April 2026 WEO severe scenario, which assumes a larger and more persistent commodity price shock. Horizontal lines indicate the cross-country average under each scenario.

Figure 6 also illustrates the increase in mortgage debt at risk held by overburdened households. The rise is substantial, with the share increasing from 16.3 percent to 21.1 percent under the FSAP geopolitical scenario, while exceeding 50 percent in Cyprus. However, the deterioration in household financial conditions associated with the Middle East war scenario is not fully captured by the conventional DSTI ratio, as the measure does not reflect the impact of rising living costs on debt affordability. The results also have potential implications for macroprudential policy. In countries experiencing the largest increases in mortgage debt at risk, policymakers may wish to consider the introduction or tightening of borrower-based measures, such as DSTI limits, where such tools are not already in place.

Once hard-to-compress essential consumption is factored in the DSECTI ratio (including food, energy, and rents), household vulnerability levels are considerably higher to start with, and rise further under stress (Figure 7). About 15 percent of euro area households could be overburdened in the baseline in 2026, compared to 3 percent on the simpler DSTI metric. In the Middle East war scenario, the share of overburdened households jumps to about one third of households, indicating that surging essential costs push many additional families into financial stress beyond those flagged by debt payments alone.

Figure 8. Households at Risk by Tenure—Cost-Adjusted Measure of Financial Stress



Sources: 2021 HFCS survey and authors' calculations. Note: The black line shows country-level aggregates of household projections for 2026 under the Baseline scenario, based on the October 2025 WEO baseline forecast. The dots represent alternative scenarios: (i) "FSAP geo", corresponding to the geopolitical scenario developed for the 2025 Euro Area FSAP; and (ii) "Middle East War", aligned with the April 2026 WEO severe scenario, which assumes a larger and more persistent commodity price shock. Horizontal lines indicate the cross-country average under each scenario.

Lower-income households bear a disproportionate brunt of the cost-of-living and interest rate shocks, despite often carrying less debt in absolute terms. Pre-shock data show a stark affordability gap by income level: in 2021, the average share of overburdened households in the bottom income tercile was under 40 percent, versus under 1 percent for the top tercile. Consequently, small financial hits can push low-income borrowers into distress. The stress projections confirm this: the average share of vulnerable households surges among the poorest tercile under all scenarios, reaching over two thirds of households in the Middle East war scenario, while high-income households stay at 5 percent even under the worst conditions.

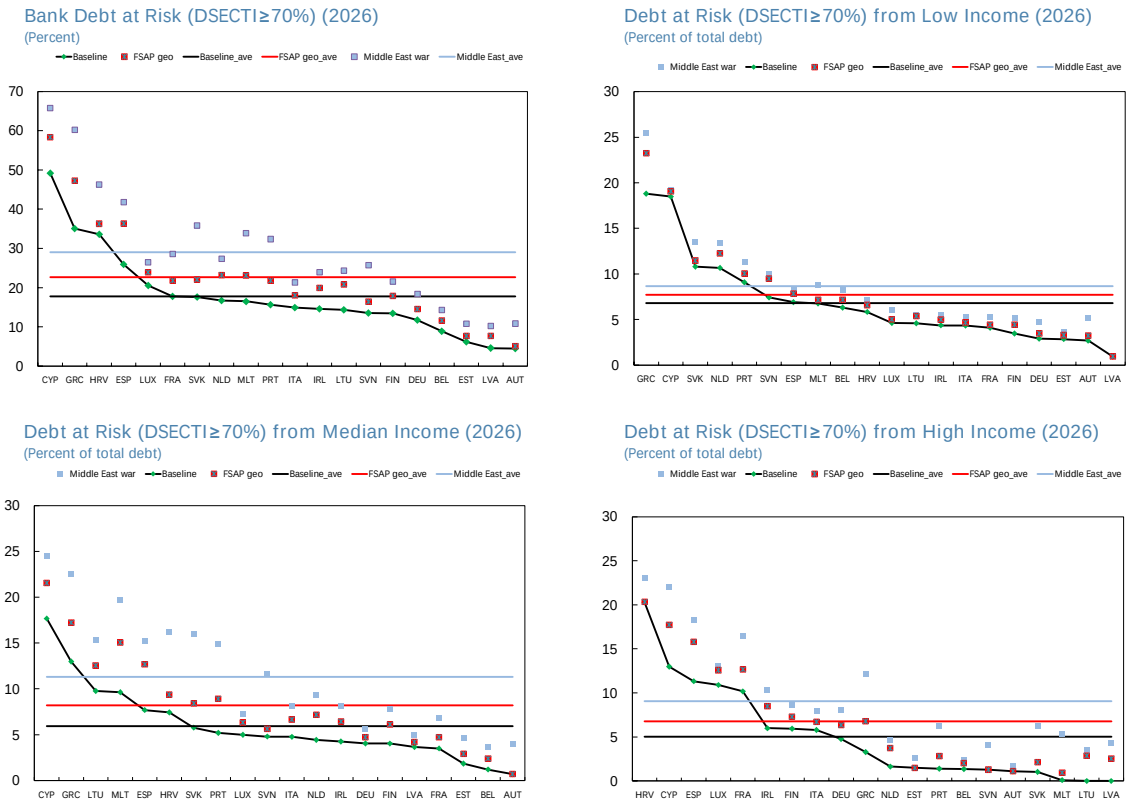
The rise in at-risk households is highly uneven across countries, reflecting different initial debt levels, shock sensitivities, income levels, mortgage contract structures, and existing borrower-based macroprudential requirements. These institutional features may partly explain the heterogeneity, although they are themselves shaped by country-specific risk conditions. Also, the Middle East war scenario stretches household finances in many Southern and Eastern European economies, including Cyprus, Croatia, Spain, Greece, and Malta where over 40 percent of households would be financially stressed. On average, renters are almost twice as likely to be overburdened than mortgage owners (Figure 8). One reason is that low-income families are more likely to rent than median- and high-income households. However, adverse

financial conditions can deteriorate households with mortgage at a faster pace, particularly in countries with floating-rate mortgages (e.g., Spain).

Debt at Risk

The implications for financial stability are illustrated in Figure 9. On average, the share of *debt at risk*—defined as household bank debt held by vulnerable borrowers—increases by 11 percentage points relative to baseline (60 percent) under the Middle East war scenario, exceeding one quarter of total outstanding debt. In Cyprus, Greece, Croatia and Spain, the share of risky debt would surpass 40 percent of bank debt in the Middle East war scenario, reflecting a combination of high leverage, the prevalence of floating-rate loans, or legacy non-performing exposures, including from the Covid-19 crisis.

Figure 9. Bank Debt at Risk by Income Level—Cost-Adjusted Measure of Financial Stress



Sources: 2021 HFCS survey and authors' calculations. Note: The black line shows country-level aggregates of bank retail portfolio projections for 2026 under the Baseline scenario, based on the October 2025 WEO baseline forecast. The dots represent alternative scenarios: (i) "FSAP geo", corresponding to the geopolitical scenario developed for the 2025 Euro Area FSAP; and (ii) "Middle East War", aligned with the April 2026 WEO severe scenario, which assumes a larger and more persistent commodity price shock. Horizontal lines indicate the cross-country average under each scenario.

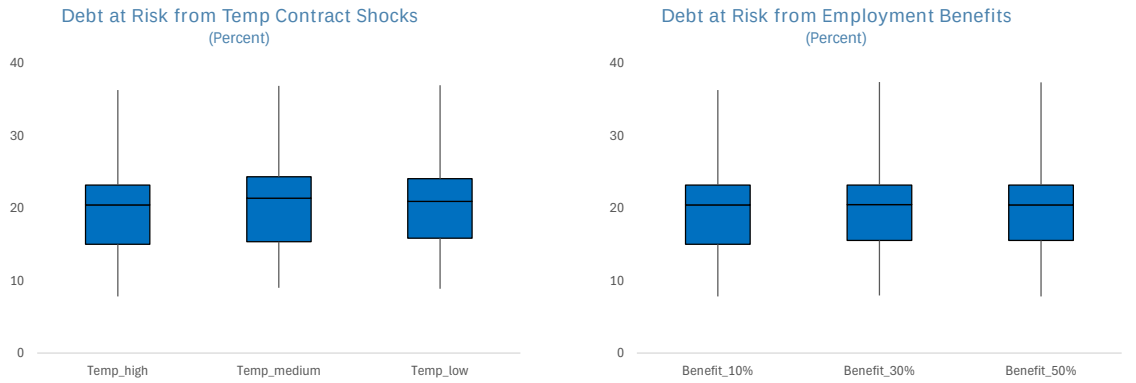
Although low-income households are more vulnerable to shocks, middle- and high-income households account for a larger share of outstanding debt. As a result, their contribution to aggregate debt at risk is greater. Under the Middle East war scenario, the increase in debt at risk is driven primarily by middle-income households (5 percentage points), followed by high-income households (4 percentage points) while low-income households contribute 2 percentage points.

Notably, there is considerable cross-country heterogeneity. In Portugal, low-income households account for nearly half of debt at risk under the Middle East war scenario, compared with 40 percent in the average country. In contrast, in Spain, high-income households account for 45 percent of debt at risk, relative to 30 percent in the average country. This heterogeneity suggests that financial-stability-oriented household support policies may need to be tailored to country-specific vulnerabilities.

Results are robust to assumptions regarding both the distribution of unemployment shocks across permanent and fixed-term contracts and the level of unemployment benefits. Figure 10 illustrates the share of debt at risk under the FSAP geopolitical scenario in 2026 for alternative assumptions about the relative severity of shocks affecting temporary versus permanent workers, as well as different levels of unemployment benefits.

While Figures 6-9 show the projections of households at risk and bank debt at risk at end 2026, similar charts are obtained for 2027 and 2028. Although the quantitative estimates differ somewhat, the order of magnitude is similar. Results are available upon request.

Figure 10. Robustness of Debt-at-Risk Estimates to Labor Market Assumptions



Sources: 2021 HFCS survey and authors' calculations. Note: The boxplots in the left panel show the cross-country distribution of debt at risk under the 2026 FSAP geopolitical scenario for alternative assumptions regarding the distribution of unemployment shocks across temporary and permanent labor contracts. Temp_high assumes that workers on temporary contracts are 11 times more likely to experience unemployment shocks than workers on permanent contracts; Temp_medium assumes they are 4 times more likely; and Temp_low assumes equal likelihood. The boxplots in the right panel show the cross-country distribution of debt at risk under the 2026 FSAP geopolitical scenario for alternative assumptions regarding unemployment benefits, expressed as a share of pre-shock labor income (10, 30, and 50 percent, respectively).

Impact on Euro Area Banks' Solvency

We next evaluate the implications of household defaults for euro area banks' capital. Specifically, we project the migration of household loans across IFRS 9 impairment stages under each scenario and estimate the associated expected credit loss provisions. The resulting projections quantify the model-implied impact of household distress on bank credit-loss provisions and regulatory capital. They do not constitute a comprehensive bank stress test because net interest income, corporate and SME exposures, market risk, management actions, diversification, and endogenous credit-supply responses are outside the framework. The reported CET1 changes isolate additional provisions on household loans, holding other capital drivers outside the exercise; they are not the full net effect of the scenario on solvency. In particular, higher rates may also raise interest income on floating-rate loans, although funding costs and other profitability effects can offset that benefit.

Procedure

Under the internationally adopted IFRS 9 accounting framework, banks classify credit exposures into three stages under the expected credit loss (ECL) approach: Stage 1 (S1), comprising performing exposures; Stage 2 (S2), comprising exposures that have experienced a significant increase in credit risk since origination but remain performing; and Stage 3 (S3), comprising non-performing exposures. While Stage 1 exposures are subject to 12-month ECL provisions, both Stage 2 and Stage 3 exposures require the recognition of lifetime ECLs. In practice, arrears are commonly used as a proxy for identifying Stage 3 exposures.

The framework developed in this paper maps projected household financials to IFRS 9 credit stages in order to assess the implications of household distress for bank asset quality. Under the benchmark IFRS 9 mapping, households are classified as follows: (i) Stage 1, households that are neither in arrears nor financially strained, defined as having a DSECTI ratio below 70 percent; (ii) Stage 2, households that remain current on their obligations but exhibit signs of financial stress, defined as having a DSECTI ratio above 70 percent; and (iii) Stage 3, households in arrears, irrespective of their financial ratios. This mapping provides a bridge between micro-level household balance sheet indicators and the accounting credit risk metrics used by banks. The estimated PDs represent the average predicted probabilities from the logistic regressions for representative households in the overburdened and non-overburdened groups, respectively.

In credit risk analysis, transition rates are defined as the likelihood a borrower will move from one IFRS 9 stage to another (e.g. from S1 to S2 and S3). We estimate these migration probabilities using projected changes in household financial vulnerability and default risk under alternative macrofinancial scenarios. Household-level projections are then aggregated to generate stage-specific transition rates over the forecast horizon. Using supervisory data from the 2023 EU-wide stress test, we calibrate these migration matrices in the household loan portfolio across IFRS 9 stages from the survey through 2028. Because all banks supervised by the ECB report the distribution of household exposures by IFRS 9 stage, together with the corresponding stock of provisions, it is possible to derive stage-specific provisioning coverage ratios for household exposures at the individual bank level. The survey does not identify lenders. We therefore treat the debt-weighted household loan sample in each country as representative of each bank's household portfolio and apply common country-level transition rates to banks' exposures in that country. Bank-specific differences arise from initial staging, provisioning coverage, exposure amounts and RWAs, not from observed differences in borrower LTV, DSTI or mortgage repricing risk. Results are reported at the aggregate bank-country level rather than as assessments of individual banks. This is a tractable approximation for aggregate analysis when portfolios are broadly similar, but aggregation does not eliminate bias: substantial within-country heterogeneity, particularly when correlated with bank size, coverage ratios or borrower risk, can affect aggregate losses as well as understate their dispersion. Portfolio-specific supervisory or suitable credit-register data would allow a more granular mapping.

The coverage ratio, defined as loan loss provisions to total exposure, is calibrated using bank-reported supervisory data as of 2023 and is assumed to remain constant throughout the stress testing horizon. As household exposures migrate from Stage 1 to Stages 2 and 3, banks are required to recognize substantially higher expected credit losses and, consequently, hold larger provisions. In 2023, the average coverage ratio reported by European banks was approximately 0.2 percent for Stage 1 exposures, 4 percent for Stage 2 exposures, and 45 percent for Stage 3 exposures. Applying these stage-specific coverage ratios to the projected migration of household exposures enables us to estimate additional provisioning needs, expected credit losses, and the resulting impact on banks' capital positions stemming from a deterioration in household credit quality under adverse macrofinancial scenarios.

Findings

We simulate the share of bank debt at risk and the transition of household exposures across IFRS 9 stages under the seven adverse macro-financial scenarios in 2026 (including the Middle East war scenario) and six scenarios over 2027-2028. Results are shown under the preferred specification (Model 4), which delivers the best in-sample fit and predictive performance. Figure 11 reports the results for the baseline, FSAP geopolitical, and Middle East war scenarios for the average euro area country.

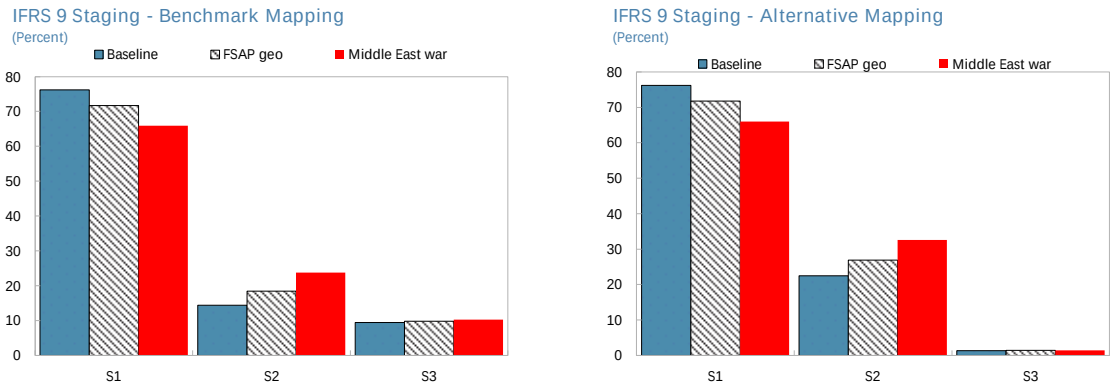
Under the benchmark IFRS 9 mapping, the share of Stage 2 loans increases from 14.4 percent in the baseline scenario to 18.4 percent under the FSAP geopolitical scenario and to 23.8 percent under the Middle East war scenario. At the same time, the share of Stage 3 loans rises by approximately 9 percent, corresponding to about 1 percentage point, relative to the baseline under the Middle East war scenario.

The results remain robust under an alternative IFRS 9 mapping. In this specification, households are classified as follows: (i) Stage 1, households that are neither in arrears nor financially strained, defined as having a DSECTI ratio below 70 percent; (ii) Stage 2, households that exhibit signs of financial stress but have arrears of less than 90 days, as well as households that are not financially strained but have arrears between 0 and 90 days; and (iii) Stage 3, households with arrears exceeding 90 days, irrespective of their financial ratios. Under this alternative mapping, the share of Stage 2 loans increases to almost one-third of total household exposures in the average euro area country under the Middle East war scenario, while the share of Stage 3 loans rises by approximately 10 percent, corresponding to around 20 basis points, relative to the baseline scenario (Figure 11).

Under the benchmark IFRS 9 mapping, the average depletion of Common Equity Tier 1 (CET1) capital is around 25 basis points under the baseline scenario in 2026. The largest capital impacts are observed under the Middle East war scenario, which results in an average CET1 depletion of around 125 basis points (Figure 12). This scenario combined a sharp increase in commodity prices, tighter financial conditions, and a decline in real disposable income. These channels reinforce each other, generating a marked deterioration in household creditworthiness and higher provisioning requirements. The results are robust across model specifications and calibration choices, with only limited quantitative variation, lending confidence to the estimated sensitivity of household credit risk to macro-financial stress. Other adverse scenarios producing sizeable capital impacts include the *combined*, *cost-of-living*, and FSAP geopolitical scenarios, under which aggregate CET1 depletion ranges between 75 and 90 basis points.

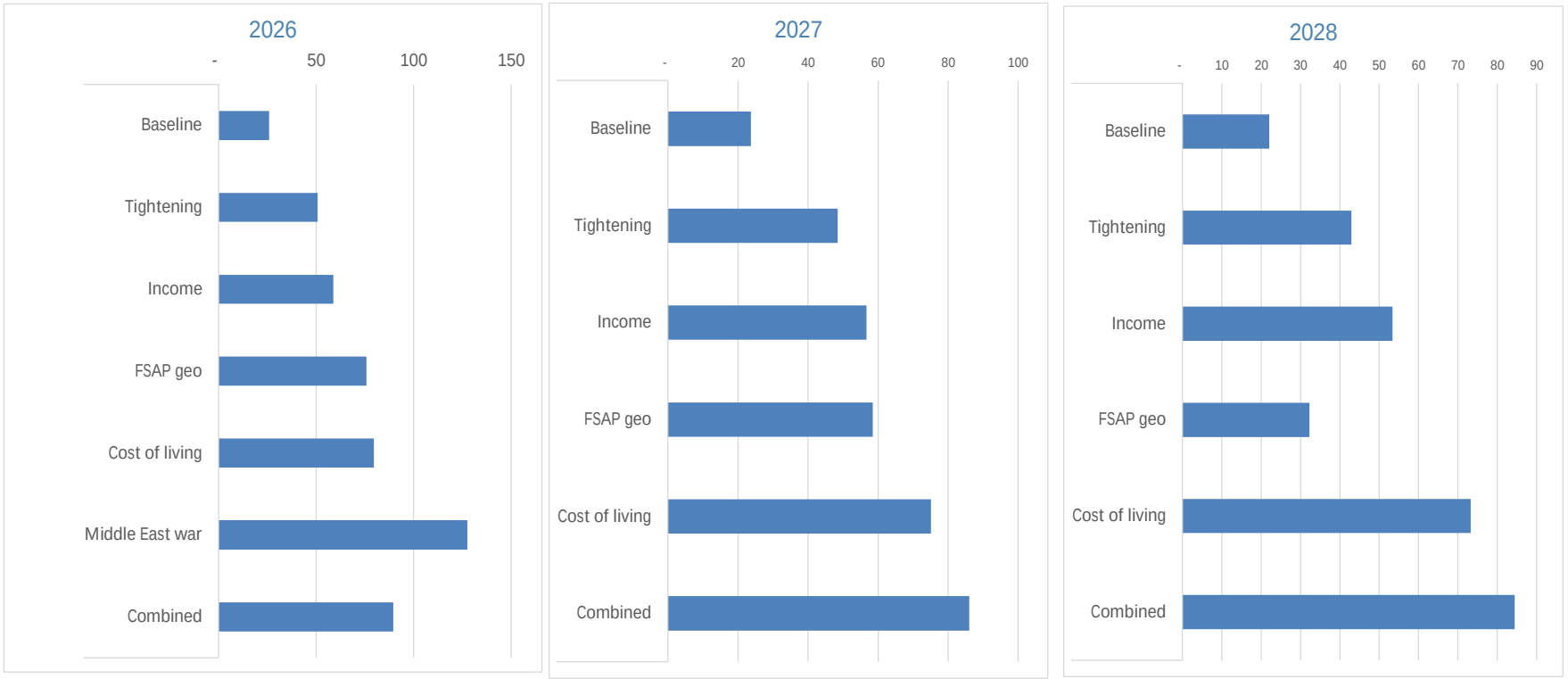
Aggregate results, however, conceal substantial cross-country heterogeneity, particularly under the Middle East war scenario (Figure 13). In a subset of countries, the cumulative impact on bank capital is considerably larger, with CET1 depletion reaching up to 200 basis points. These differences highlight the importance of country-specific household balance sheet characteristics, mortgage market structures, and institutional features in shaping stress-test outcomes. Capital impacts are also markedly front-loaded, with most losses materializing in the first year of the stress horizon. The effect moderates in subsequent years, reflecting the assumed macroeconomic recovery embedded in the scenarios and the gradual repair of household balance sheets over time.

Figure 11. Robustness of IFRS 9 Staging to Mapping Assumptions



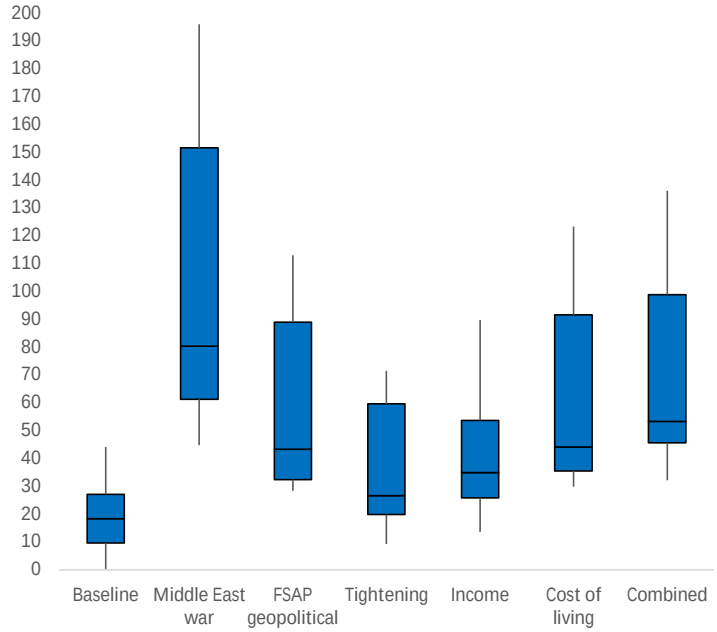
Sources: 2021 HFCS survey and authors' calculations. Note: The left (right) panel shows the distribution of household loans across IFRS 9 stages under the benchmark (alternative) mapping and the baseline, FSAP geopolitical, and Middle East war scenarios. Under the benchmark mapping, Stage 1 includes households that are neither in arrears nor financially strained (DSECTI < 70 percent), Stage 2 includes financially strained households that remain current on their obligations (DSECTI ≥ 70 percent), and Stage 3 includes all households in arrears. Under the alternative mapping, Stage 2 includes financially strained households with arrears of less than 90 days and non-strained households with arrears of 0-90 days, while Stage 3 includes households with arrears exceeding 90 days, irrespective of their financial condition.

Figure 12. CET1 Depletion Across Scenarios and Stress Test Horizon in the Euro Area
(Basis points)



Source: 2021 HFCS, 2023 EBA supervisory data; and authors' calculations. Note: The panels show the average bank capital depletion from higher loan loss provisions in the household loan portfolio, measured as the decline in Common Equity Tier 1 regulatory capital (CET1) expressed in basis points of risk weighted assets (RWA), averaged across euro area countries. Results are shown for the "Model 4" specification estimated using the pooled cross-sectional logit regression. The scenarios considered are "Baseline", "Tightening", "Income", "FSAP geo", "Cost of living", "Middle East war" (2026) and "Combined".

Figure 13. CET1 Depletion Across Countries by Scenario, 2026
(Basis points)

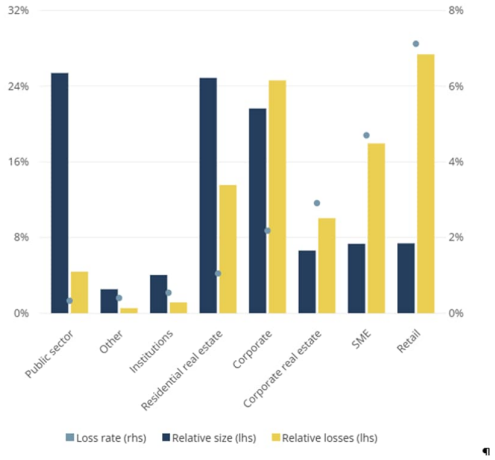


Source: 2021 HFCS, EBA supervisory data; and authors' calculations. Note: The box plot shows the cross-country distribution of the bank capital depletion from higher loan loss provisions in the household loan portfolio, measured as the decline in Common Equity Tier 1 regulatory capital (CET1) expressed in basis points of risk weighted assets (RWA). Results are shown for the "Model 4" specification estimated using the pooled cross-sectional logit regression. The scenarios considered are "Baseline", "Tightening", "Income", "FSAP geo", "Cost of living", "Middle East war" and "Combined".

We benchmark our results against the 2025 EBA EU-wide stress test, focusing on the credit risk channel (Figure 14). While the EBA exercise incorporates a comprehensive set of risk drivers—including market risk and the impact of net interest income—our framework isolates the credit risk component arising from households' repayment capacity under adverse macroeconomic conditions. Consequently, the estimated capital effects should be interpreted as a partial measure of stress, capturing only the deterioration in asset quality and the associated increase in provisioning requirements on the retail portfolio.

In comparison, the EBA stress test identifies credit risk as the dominant source of losses, particularly in retail and SME portfolios, with materially lower loss rates for the public sector and other low-risk exposures. Although our model does not cover the full balance sheet and excludes offsetting income effects, the relative importance of household-related exposures in driving capital depletion is consistent with the EBA evidence. The main difference lies in scope: whereas the EBA framework aggregates losses across corporate, real estate and other portfolios, our approach concentrates on the household segment and provides a granular assessment of this specific transmission channel. Within this narrower domain, the order of magnitude of capital erosion under severe scenarios is broadly aligned with the credit risk contribution observed in the EBA exercise, albeit without the mitigating effects stemming from diversification and income generation included in the supervisory framework.

Figure 14. Credit Risk Losses of EU banks in the 2025 EU-wide Stress Test
(Percent)



Source: EBA https://www.eba.europa.eu/publications-and-media/press-releases/eba-publishes-results-its-2025-eu-wide-stress-test Note: The chart shows the breakdown of credit risk losses in the EBA 2025 stress test. Household-related exposures are captured under "Residential real estate" and "Retail."

Impact on Consumption

The household finance literature identifies labor income shocks, employment transitions, interest-rate changes, cost-of-living pressures, wealth fluctuations, liquidity constraints, and uncertainty as the primary drivers of household consumption (Crawley and Theloudis, 2024). The magnitude of consumption responses varies considerably across households depending on their leverage, savings, credit access, and marginal propensity to consume. As private consumption accounts for more than half of GDP in the euro area, changes in household spending can impact economic activity. Lower-income families spend a much larger fraction of income on essentials (about 70 percent for the bottom tercile, versus 34 percent for middle-income) and typically exhibit a higher marginal propensity to consume. By contrast, middle-income families tend to be more leveraged as they are more likely to hold mortgages. Thus, a commodity shock combined with higher interest rates that is not offset by wage growth can eliminate any discretionary spending across a broad range of households.

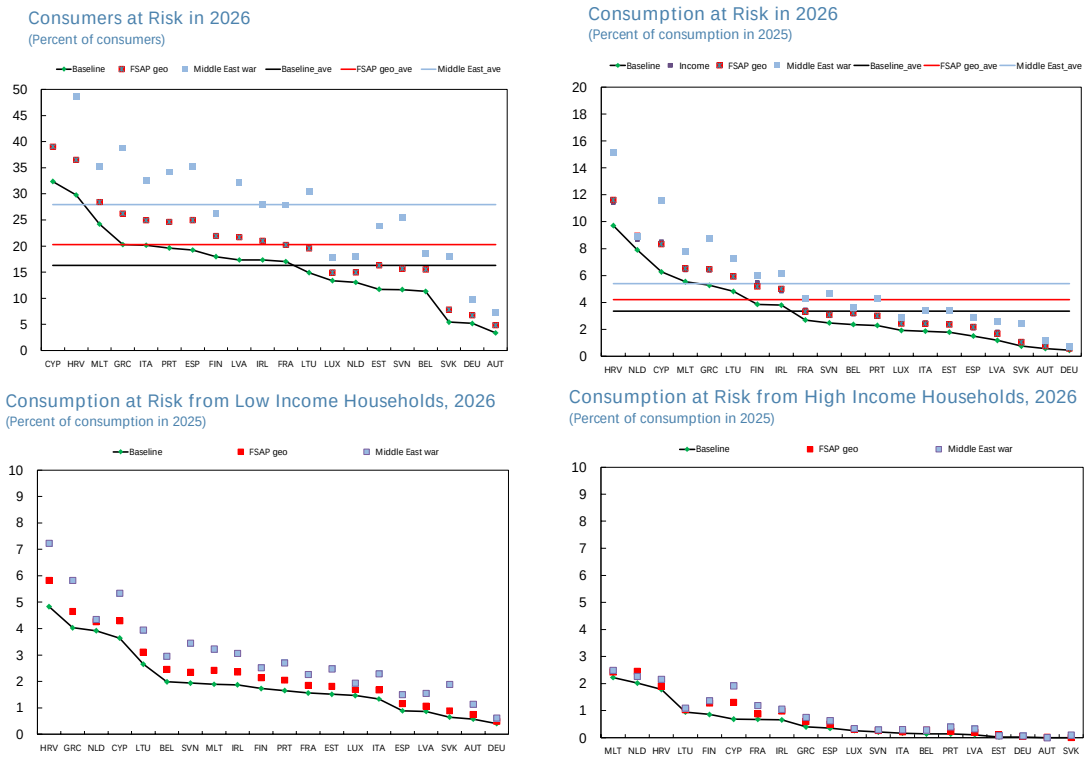
To assess the macroeconomic implications of the consumption channel under adverse conditions, we consider an alternative measure of household vulnerability. A household is classified as a *consumer-at-risk*, i.e., at an elevated risk of cutting consumption, if debt service, rents, and total consumption exceed gross income (DSTCTI>1). Formally, the definition of DSTCTI is shown in equation (4).

$$DSCTI^{h}_{T,j} = DSECTI^{h}_{T,j} + \frac{other\ consumption^{h}_{t} \cdot \left(1 + \Delta other\ consumption^{h}_{T-t,j}\right)}{I^{h}_{t} \cdot \left(1 + \Delta I^{h}_{T-t,j}\right)} \quad \text{.....(4)}$$

where h denotes the household; $DSECTI$ is the debt service and essential consumption (including rents) to income ratio; $other\ consumption$ is the amount spent on discretionary consumption; and I is total gross household income. It is important to note that the measure of consumer-at-risk differs from the definition of household-at-risk, as it also takes into account non-essential consumption.

Households for which debt service, rents, and total consumption exceed gross income face a binding budget constraint and may be forced to cut spending. We assume that these households can partially smooth consumption by drawing down their savings. Given the heightened uncertainty, households are allowed to deplete up to half of their savings during the first year of stress, 2026. While some households may rely on additional borrowing or credit lines to further smooth consumption, this channel is likely to be constrained by the high cost of credit. Conversely, other households may adjust their spending well before the budget constraint becomes binding for precautionary reasons. We expect these opposing effects to broadly offset each other on average. Because taxes and social contributions are not deducted, this indicator can miss households whose spending exceeds disposable income but remains below gross income. It is not a direct measure of a binding after-tax budget constraint. The assumption that precautionary reductions and additional borrowing offset on average is not separately estimated.

Figure 15. Consumption at Risk

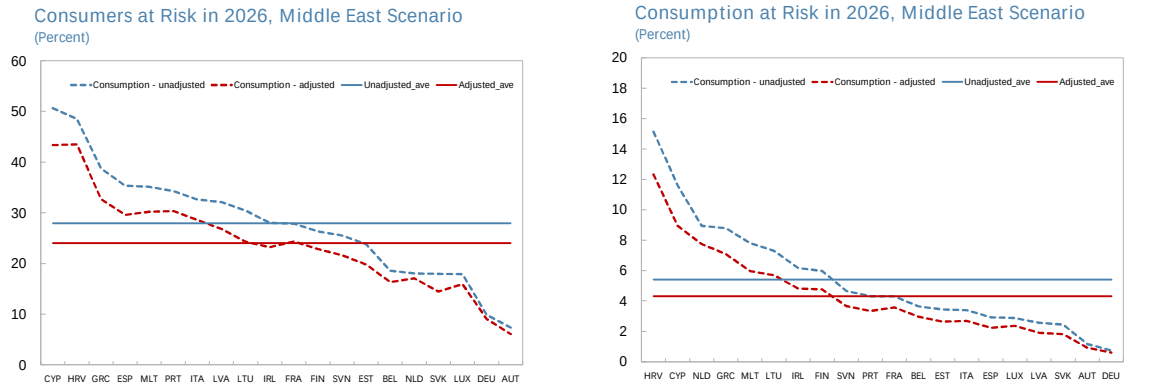


Sources: 2021 HFCS survey and authors' calculations. Note: The black line shows country-level aggregates of household projections for 2026 under the Baseline scenario, based on the October 2025 WEO baseline forecast. The dots represent alternative scenarios: (i) "FSAP geo", corresponding to the geopolitical scenario developed for the 2025 Euro Area FSAP; and (ii) "Middle East War", aligned with the April 2026 WEO severe scenario, which assumes a larger and more persistent commodity price shock. Horizontal lines indicate the cross-country average under each scenario.

We define *consumption-at-risk* as the reduction in consumption required for *consumers-at-risk* to satisfy their budget constraint after drawing down their liquid financial assets, including cash, deposits, and mutual fund holdings, up to one-half of available savings in 2026. This measure is calculated at the household level and then aggregated using household population weights. The numerator is the sum of each affected household's required reduction in non-essential consumption after the permitted savings drawdown, and the result is expressed relative to aggregate 2025 consumption. Consumption-at-risk is expressed as a share of aggregate consumption in 2025. We assume that households adjust only non-essential expenditure.

The impact of the Middle East crisis on consumption depends on whether households adjust their energy consumption to mitigate higher energy costs. We consider two alternative assumptions. Under the "*no behavioral adjustment*" assumption, real consumption in 2026 follows the WEO projections for each country. Under the "*behavioral adjustment*" assumption, households reduce their consumption of energy, including electricity, gas, and other fuels, in line with the average adjustment observed in the euro area during the 2022 energy crisis, equivalent to a year-on-year decline of about 6 percent in real terms. The uniform 6 percent adjustment is an illustrative sensitivity based on an aggregate historical response, not an estimated household- or country-specific price elasticity. It does not allow for differences in shock size, climate, housing efficiency, energy mix or ability to reduce essential energy use, and should not be interpreted as a behavioral forecast.

Figure 16. Sensitivity of Consumption to Behavioral Responses in Energy Use



Sources: 2021 HFCS; Haver; and authors' calculations. Note: The left (right) panel reports country-level estimates of the share of consumers-at-risk (consumption-at-risk) under two assumptions: (i) households' real energy consumption evolves in line with WEO projections (consumption-unadjusted, red line), and (ii) households adjust energy consumption following the pattern observed during the 2022 energy crisis (consumption-adjusted, blue line). Horizontal lines denote the euro area average across countries.

Our results suggest that, under the Middle East war scenario and assuming no behavioral adjustment, nearly 30 percent of households would be forced to reduce consumption by 2026. On average, approximately 5.4 percent of aggregate consumption, relative to its estimated 2025 level,[8] could be at risk as essential payments and discretionary spending exceed households' income and available savings. These results are derived under a no-policy scenario. For comparison, aggregate consumption in the euro area contracted by 8.1 percent during the COVID-19 pandemic, while consumption increased during the 2022 energy crisis, largely reflecting extensive policy support measures. In two countries, the contraction exceeds 10 percent of aggregate consumption (Figure 15). Low-income households account for the largest share of the decline, contributing 56 percent of the reduction in aggregate consumption on average across countries, compared with 29 percent for middle-income households and 15 percent for high-income households. Nevertheless, substantial cross-country heterogeneity remains. In Austria, Belgium, Germany, and Slovakia, more than 75 percent of the decline originates from low-income households, whereas in France, Malta, and the Netherlands, high-income households account for more than one-quarter of the total reduction.

When households adjust their energy consumption in line with observed behavior during the 2022 energy crisis, the share of households at risk falls to 25 percent, while aggregate consumption at risk declines to about 4.3 percent relative to 2025 levels. Under this assumption, the contraction in consumption exceeds 10 percent only in Croatia. Behavioral adjustment has the largest effect in Croatia and Cyprus, where consumption at risk declines by about 3 percentage points relative to the no-adjustment scenario (Figure

[8] This level is estimated using microdata on consumption of goods and services from the 2021 HFCS survey, scaling for capital expenditure and adjusted using the country-level growth rate of nominal private consumption between 2021 and 2025.

16). These findings underscore the importance of targeted policy interventions to support vulnerable households and mitigate adverse spillovers to the real economy. Household financial distress and macroeconomic weakness are closely interconnected: financially stretched households are not only more likely to default on their debt obligations but are also more likely to reduce consumption in order to remain current on those obligations, amplifying the transmission of adverse shocks to economic activity.

The Impact of Policies

European Fiscal Response to the Middle East War

The fiscal measures adopted in response to the oil price surge from the Middle East war were, for the most part, broad-based rather than targeted. The IMF estimates that around 80 percent of total support in the euro area took the form of untargeted tax cuts and price-suppressing measures, with only a limited number of countries introducing relief aimed specifically at vulnerable households. This composition marks a departure from past cost-of-living episodes (1996–2025), when advanced economies leaned more heavily on income-supporting instruments—around 60 percent of measures—half of which were targeted by income or age (IMF, 2026b). The European Commission reached a similar assessment, noting that roughly three-quarters of announced measures were non-targeted. It cautioned that support should be temporary, targeted, and designed to preserve incentives to reduce energy consumption. Beyond tax relief, governments deployed a range of compensatory instruments—direct income transfers, energy vouchers, heating subsidies, and support for energy-intensive sectors.

Table 4. European Fiscal Response to Households during the Middle East War

Type of measure	Measure	Description	Examples
Income-supporting measures	Transfers (targeted)	Lump-sum payments or targeted cash transfers to households to offset higher energy costs.	Targeted compensation for low/income vulnerable groups (e.g., Croatia, Greece)
	Transfers (untargeted)	Lump-sum payments or cash transfers to all consumers to offset higher energy costs.	One-off household payments based on actual Jan-Feb 2026 electricity and gas consumption (e.g., Sweden)
Price-supporting measures	VAT reductions	Temporary reductions in VAT rates on electricity, gas, transport fuels, or heating products.	Spain's package included VAT reductions on fuel, electricity, and gas.
	Excise duty reductions	Temporary cuts in fuel and energy excise taxes to lower retail prices.	Germany's energy tax cut; Sweden's vehicle fuel duty reduction.
	Fuel rebates	Subsidies or rebates aimed at reducing transport fuel costs for households and firms.	Sectoral fuel rebates in several European countries.
	Electricity and gas price compensation	Government-funded compensation to offset increases in utility bills.	Sweden's electricity and gas compensation scheme.
	Reduced electricity taxes	Targeted tax relief designed to lower electricity costs while focusing support on vulnerable consumers.	Included in the Commission's AccelerateEU toolbox.

Sources: Bruegel (2026 European Energy Crisis Fiscal Response Tracker); European Commission (2026 Policy Measures in EU Member States to Address the Energy Price Shock; AccelerateEU Communication), and authors' calculations.

Evidence from the IMF Global Policy Tracker suggests that the composition of policy responses to the economic impact of the Middle East war closely resembles that observed during the 2022 energy crisis, although fiscal space is generally more constrained due to higher public debt and rising interest costs. Internal IMF analyses indicate that, by April 2026, the tracker had recorded 645 policy measures across 146 countries, with advanced European economies relying mainly on energy subsidies, excise tax reductions, and other fiscal measures to cushion households against the energy shock.

While these measures helped protect purchasing power and limit the short-term impact on inflation and household distress, Bruegel, the Commission and the IMF cautioned that broad-based subsidies can be fiscally costly and less efficient than targeted transfers to vulnerable households. Table 4 provides an overview of the range of fiscal measures introduced by selected European governments to mitigate the impact of the Middle East war energy crisis on European households. Additional measures were implemented to support European firms.

Cost Effectiveness of Household Transfers

This section evaluates the cost-effectiveness of household transfers under alternative targeting schemes using a simulation-based approach. The transfer takes the form of a lump-sum cash payment designed to offset part of the increase in household energy costs. Specifically, we assume that the transfer compensates households for one-half of the rise in energy expenditures projected for 2026 relative to end-2025 levels. Using household-level information on utility expenditures from the 2021 HFCS, we project households' utilities spending in 2026 by applying the cumulative increase in the country-specific energy component of the Harmonized Index of Consumer Prices (HICP) from the relevant survey reference date to end-2025 and then under the 2026 scenario. Combining these estimates with information on the number of households in each income group, we calculate the fiscal cost of absorbing one-half of the increase in energy prices. Relative to the full-pass-through no-policy benchmark, these transfers absorb half of the increase in eligible households' energy bills, reducing effective pass-through to their net budget burden. Retail prices and marginal price signals are not mechanically reduced by a lump-sum transfer. The policy exercise should therefore be distinguished from a model of price caps or country-specific wholesale-to-retail pass-through

We consider three alternative targeting mechanisms based on household income. Under a *broad* transfer scheme, all households receive support. Under a *partially targeted* scheme, transfers are provided to the bottom two-thirds of the income distribution. Under a *narrowly targeted* scheme, only households in the bottom income tercile receive support. We evaluate the effectiveness of each policy under the protracted Middle East war scenario.

Policy benefits are assessed using two complementary metrics. First, we measure the reduction in the share of households experiencing financial distress (a welfare measure). Second, we quantify the amount of consumption-at-risk preserved relative to a no-policy baseline (an economic measure). Expressing these consumption savings as a share of GDP allows for a direct comparison between the economic benefits and the fiscal resources required, both measured in terms of GDP.

The cost-effectiveness of transfer schemes depends on whether households adjust their energy consumption in response to higher prices under the Middle East war scenario. We therefore evaluate the effectiveness of transfers under two assumptions: "*no behavioral adjustment*", where households maintain their pre-shock energy consumption levels, and "*behavioral adjustment*", where households reduce energy consumption in response to higher prices. While the fiscal cost of transfer schemes is expected to be lower under the behavioral adjustment scenario because compensation is provided on a smaller quantity of energy consumption, the effect on policy benefits is less straightforward. Although the share of financially distressed households falls in absolute terms when households adjust their consumption, the improvement relative to the corresponding no-policy scenario may be either larger or smaller.

Under the "*no behavioral adjustment*" scenario, results indicate that the narrowly targeted transfer scheme is the most cost-effective policy for reducing household financial distress (Figure 17, left panel). On average, this policy reduces the share of financially distressed households by about 2.8 percentage points at a fiscal cost slightly above 0.5 percent of GDP under the Middle East war scenario, yielding a cost-effectiveness ratio of 5.3 percentage points of households lifted out of financial distress per percentage

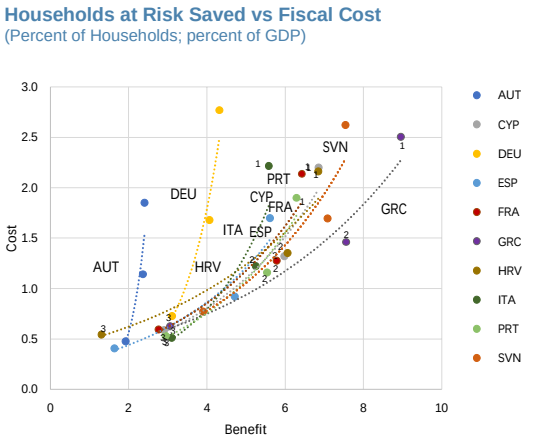
point of GDP spent. The corresponding ratio declines to 4.1 under the partially targeted scheme and to 2.8 under the broad transfer scheme. However, the effectiveness of targeting varies considerably across countries.

Under the "*behavioral adjustment*" scenario, the narrowly targeted transfer scheme remains the most cost-effective policy (Figure 17, right panel). On average, it reduces the share of financially distressed households by about 2.9 percentage points at a fiscal cost slightly below 0.5 percent of GDP, resulting in a cost-effectiveness ratio of 5.9 percentage points of households lifted out of financial distress per percentage point of GDP spent. The corresponding ratio falls to 4.3 under the partially targeted scheme and to 2.8 under the broad transfer scheme.

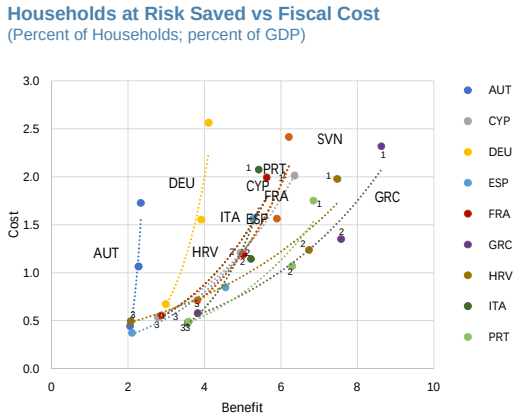
However, the effectiveness of targeting differs significantly across countries. The cost-benefit curves in Figure 17 are flatter in several Southern and Emerging European economies, including Croatia, Greece, and Slovenia, indicating that extending support beyond low-income households generates relatively large additional benefits. In these countries, middle-income households are more exposed to the shock because of high indebtedness and elevated housing-related expenditures. By contrast, Austria and Germany exhibit much steeper cost-benefit curves, implying that broadening support beyond vulnerable households results in rapidly increasing fiscal costs with only limited additional gains. This pattern reflects stronger household balance sheets and lower expenditure shares devoted to essential goods and services.

Figure 17. Welfare Impact of Fiscal Policies

Real Consumption of Energy Unadjusted



Real Consumption of Energy Adjusted



Sources: 2021 HFCS; Haver; and authors' calculations. Note: Each curve represents a country-specific cost-benefit frontier for income support policies under the Middle East War scenario. Policy benefits are measured as the share of households prevented from falling into financial distress, while policy costs are measured as the associated fiscal expenditure as a share of GDP. Three transfer schemes are considered: (1) *Broad*, under which all households receive support; (2) *Partially Targeted*, under which support is provided to households in the bottom two income terciles; and (3) *Narrowly Targeted*, under which only households in the bottom income tercile receive support. The left panel assumes that households do not adjust their energy consumption in 2026, whereas the right panel assumes that households reduce energy consumption by the same average magnitude observed in the euro area during the 2022 energy crisis. A steeper cost-benefit frontier indicates higher policy cost per household protected or consumption saved and is observed in several core economies (e.g., Austria and Germany), whereas flatter frontiers are observed in some southern economies (e.g., Croatia, Greece, Portugal, and Slovenia).

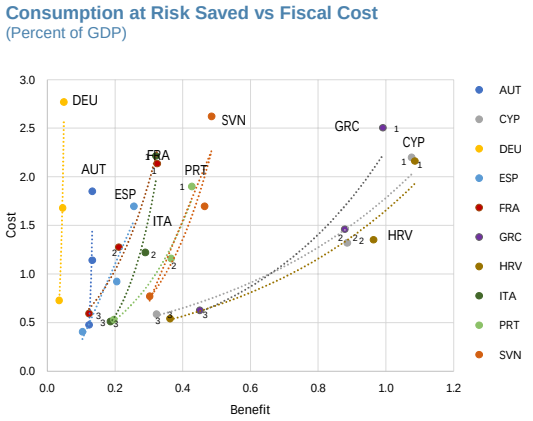
When benefits are measured instead by the amount of consumption-at-risk preserved relative to fiscal costs, none of the transfer schemes fully pays for itself in terms of higher aggregate consumption (Figure 18). Under the "*no behavioral adjustment*" scenario, the consumption-based cost-effectiveness ratio is approximately 0.4 for narrowly targeted transfers, 0.3 for partially targeted transfers, and about 0.2 for broad transfers. Under the "*behavioral adjustment*" scenario, the ratio declines slightly to around 0.3 for narrowly targeted transfers and 0.2 for both partially targeted and broad transfers. In other words, in all

countries and across all transfer designs, the increase in aggregate consumption generated by the transfers remains below the associated fiscal cost. Targeted transfers may nevertheless be justified on distributional or social-protection grounds, even when their measured short-run consumption benefits fall below their fiscal costs.

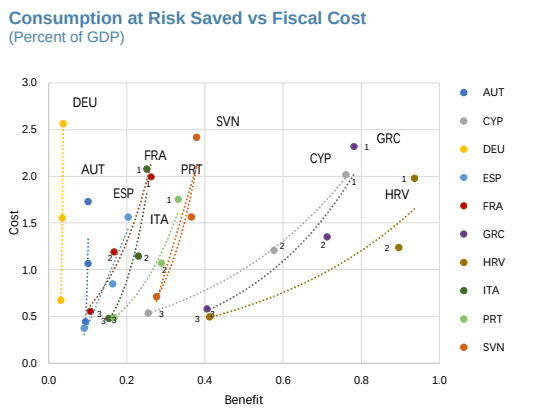
From a financial stability perspective, household transfers generate additional benefits by reducing bank losses and preserving regulatory capital. By easing pressure on household balance sheets, transfers lower the likelihood that borrowers migrate to IFRS 9 Stage 2 or Stage 3 classifications, thereby reducing provisioning needs. The results suggest that a *partially targeted* transfer scheme preserves approximately 15 basis points of bank capital on average, with impacts exceeding 30 basis points in Cyprus and Portugal (Figure 19). This is equivalent to a cost-effectiveness ratio of 14 basis points of capital preserved per percentage point of GDP spent, compared with only 3 under the *narrowly targeted* scheme. Policies directed toward the bottom two-thirds of the income distribution therefore provide the greatest financial stability benefits per unit of fiscal cost, primarily because middle-income households are more likely to hold mortgage debt and larger loan balances than lower-income households. As a result, preventing distress among these households has a disproportionately large effect on banks' credit losses and capital positions.

Figure 18. Consumption Impact of Fiscal Policies

Real Consumption of Energy Unadjusted



Real Consumption of Energy Adjusted



Sources: 2021 HFCS; Haver; and authors' calculations. Note: Each curve represents a country-specific cost-benefit frontier for income support policies under the Middle East War scenario. Policy benefits are measured as the share of consumption saved, while policy costs are measured as the associated fiscal expenditure as a share of GDP. Three transfer schemes are considered: (1) *Broad*, under which all households receive support; (2) *Partially Targeted*, under which support is provided to households in the bottom two income terciles; and (3) *Narrowly Targeted*, under which only households in the bottom income tercile receive support. The left panel assumes that households do not adjust their energy consumption in 2026, whereas the right panel assumes that households reduce energy consumption by the same average magnitude observed in the euro area during the 2022 energy crisis. A steeper cost-benefit frontier indicates higher policy cost per household protected or consumption saved and is observed in several core economies (e.g., Austria and Germany), whereas flatter frontiers are observed in some southern economies (e.g., Croatia, Cyprus, Greece, and Slovenia).

Cost Effectiveness of Interest Payments Relief

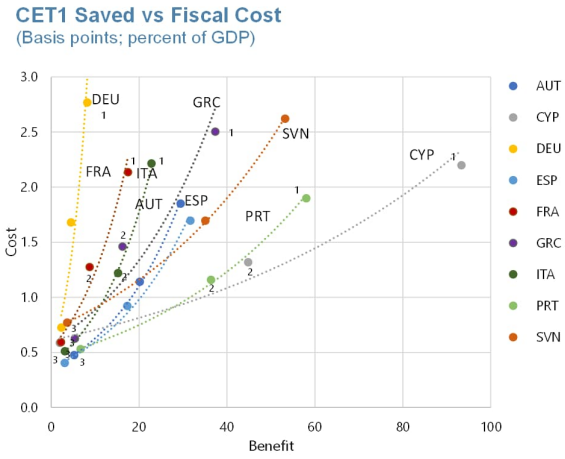
Under a protracted Middle East war scenario, characterized by tighter financial conditions and a sustained increase in the cost of living, banks may offer distressed borrowers restructuring solutions aimed at reducing monthly debt service burdens. Similar schemes were implemented in several European countries during the 2022 monetary tightening cycle, particularly in jurisdictions with a high prevalence of floating-rate mortgages. Drawing on key elements of the Portuguese mortgage relief framework, we assume that borrowers become eligible for restructuring if their DSTI ratio reaches 50 percent or more, or if their DSTI

exceeds 36 percent and increases by at least 5 percentage points in 2026. Eligibility in this simulation is evaluated using gross-income DSTI. It is inspired by, but is not a replication of, Portuguese rules defined on a different income basis; identical numerical thresholds do not imply identical eligible populations.

We consider two alternative restructuring options. Under the first option, banks defer the increase in interest payments arising in 2026 under the Middle East war scenario relative to the baseline and reschedule these payments to 2027. The benefit of this measure stems from a lower migration of loans into underperforming categories, resulting in reduced provisioning needs and a smaller impact on regulatory capital. The cost arises from the loss in the present value of interest income, since cash flows are received one year later than originally scheduled. The deferral calculation assumes full recovery of the postponed interest in 2027. It excludes additional default risk from persistent distress and the higher repayment burden when deferred amounts fall due; its low cost is therefore conditional on repayment capacity recovering, not a general finding about deferral programs.

To quantify this cost, deferred interest payments are discounted using the average interest rate on new household loans observed in March 2026. The relief applies only to loans that are eligible for restructuring and remain performing throughout 2026. Figure 20 shows substantial cross-country variation in eligibility rates, ranging from more than one-half of outstanding loans in Cyprus to around 5 percent in the Baltic countries.

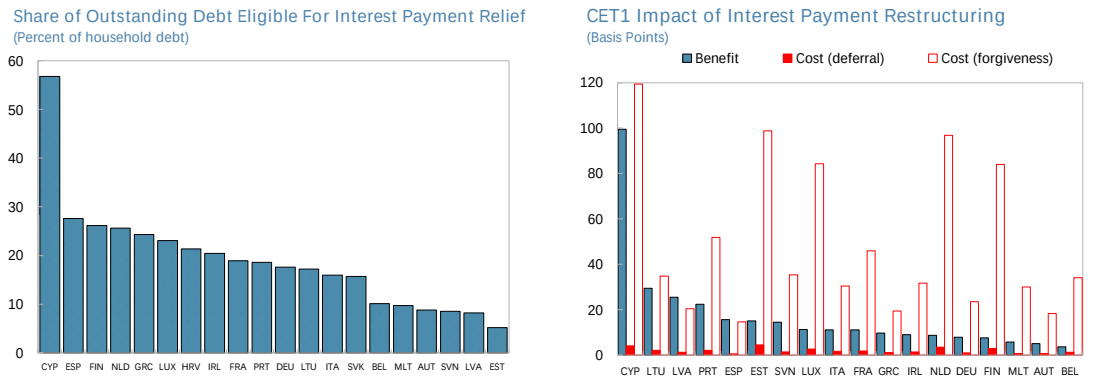
Figure 19. Bank Capital Impact of Fiscal Policies



Source: 2021 HFCS; EBA supervisory data; Haver; and authors' calculations. Note: Each curve represents a country-specific cost-benefit frontier for income support policies under the Middle East War scenario. Policy benefits are measured as the CET1 capital saved while policy costs are measured as the associated fiscal expenditure as a share of GDP. Three transfer schemes are considered: (1) *Broad*, under which all households receive support; (2) *Partially Targeted*, under which support is provided to households in the bottom two income terciles; and (3) *Narrowly Targeted*, under which only households in the bottom income tercile receive support. A steeper cost-benefit frontier indicates higher policy cost per bank capital saved and is observed in several core economies (e.g., France and Germany), whereas flatter frontiers are observed in some southern economies (e.g., Cyprus, Portugal, and Slovenia).

Under the second option, banks forgive the increase in interest payments that would otherwise accrue in 2026 under the Middle East war scenario relative to the baseline. While this option delivers the same benefit in terms of reducing loan migration and preserving capital, its cost is considerably higher because the additional interest income on eligible loans is written off entirely rather than postponed.

Figure 20. Cost-Benefit Analysis of Interest Payment Relief



Sources: 2021 HFCS; EBA supervisory data; Haver; and authors' calculations. Note: The left panel shows the share of outstanding debt eligible for interest rate relief. Borrowers qualify for support if their DSTI exceeds 50 percent, or if their DSTI exceeds 36 percent and increases by at least 5 percentage points under the 2026 stress scenario. The right panel shows the benefits of the two policy interventions, *Cost Deferral* and *Cost Forgiveness*, measured in terms of their CET1 capital impact.

Figure 20 compares the costs and benefits of the two restructuring schemes. The capital benefit of interest-payment relief is largest in Cyprus, where it reaches approximately 100 basis points of CET1 capital, and averages around 20 basis points across countries. However, the cost of outright interest forgiveness is substantially higher, averaging close to 50 basis points of CET1 capital, whereas the cost of deferring interest payments is minimal, averaging only about 2 basis points. The results illustrate that temporary deferral of higher interest payments can reduce near-term provisioning needs at a low present-value cost. However, the exercise assumes that deferred 2026 payments are repaid in 2027 and does not model the additional credit risk created by combining original and deferred obligations; the apparent near-term benefit may therefore partly reflect a shift in the timing of losses. If borrowers remain distressed, delayed payments may not be recovered in full, and additional provisions, modification losses or other costs may arise. Deferral is not assumed to remove a significant increase in credit risk or to justify automatic IFRS 9 reclassification. The reported comparison is not a full lifetime expected-loss or accounting assessment; repayment-risk sensitivity would be needed before drawing operational policy conclusions.

Conclusions

This paper develops an integrated, microdata-based framework that links household balance sheets, credit risk, bank solvency, and consumption, and that lends itself to policy simulation. By combining the 2021 HFCS with supervisory bank-level data, the framework delivers a granular assessment of household vulnerabilities and capital impact that is unattainable with aggregate data—which mask the concentration of leverage, debt burdens, and marginal propensities to consume across the income distribution and across bank portfolios. Its central methodological contributions are fivefold: estimating household default risk from observed arrears in a cross-country setting; identifying the best predictor of default through horse-race econometrics; mapping household loans into IFRS 9 impairment stages consistent with the forward-looking provisioning framework used by banks; quantifying consumption at risk; and proposing a cost-effectiveness framework for evaluating fiscal and financial policies, applied to a protracted Middle East war scenario.

A recurring message is that *who* becomes overburdened by cost-of-living expenses and *who* holds the debt—and under what conditions—critically shapes economic and financial stability outcomes. Household vulnerability and debt are unevenly distributed across income levels, with consequences for both credit risk and macroeconomic stability. Lower-income households devote a much larger share of income to essential expenses, leaving them with thin buffers, so adverse shocks such as job loss or falling real income can rapidly translate into arrears; our estimates place default probabilities for the bottom-income tercile at nearly double those of the top tercile under stress. Yet vulnerability is not confined to the poorest: middle-

income borrowers often carry the highest leverage and substantial mortgage exposure, so widespread strain in this group also poses significant risks to banks.

A key empirical finding is that the cost-of-living-adjusted debt-service ratio (DSECTI), which augments debt-service obligations with essential consumption and rents, substantially outperforms conventional debt-service ratios in predicting household default. A threshold near 70 percent maximizes the gap in default probabilities between overburdened and non-overburdened households, and default risk more than doubles once households cross it. This suggests that affordability metrics that ignore essential expenditures materially understate household vulnerability during cost-of-living crises, when the prices of necessities rise faster than incomes—with direct implications for the design of borrower-based macroprudential tools and for supervisory monitoring of household credit risk.

Applying the framework to a protracted Middle East war scenario in line with the April 2026 WEO, reveals that aggregate resilience coexists with acute pockets of vulnerability. While the median euro area household withstands the shock, the share of overburdened households rises to roughly one-third, with low-income households experiencing the most severe strain. Because middle- and high-income households hold the bulk of outstanding debt, however, they account for a disproportionate share of the increase in debt at risk. The resulting deterioration in household credit quality translates into an illustrative average CET1 depletion from household credit-loss provisions of about 125 basis points, with substantial cross-country dispersion driven by differences in leverage, mortgage structure, and legacy asset quality. These findings reinforce the view that the distribution of debt, not merely its aggregate level, is decisive for financial stability.

The paper also documents a tight link between household financial fragility and aggregate demand. Households facing binding budget constraints curtail discretionary consumption to remain current on their debt; under the severe Middle East war scenario, nearly 30 percent become consumers-at-risk, implying an aggregate consumption contraction of about 5.4 percent relative to 2025 with no behavioral adjustment. When households adjust their energy consumption in line with patterns observed during the 2022 energy crisis, the share of consumers-at-risk falls to 25 percent and aggregate consumption-at-risk declines to 4.3 percent. More broadly, the simulations quantify two transmission channels: rising household defaults that erode bank capital through higher credit-loss provisions, and reduced household consumption that weakens aggregate demand. The framework does not incorporate general-equilibrium feedback effects from lower consumption to firms, employment, borrower incomes, bank balance sheets, or credit supply. Accordingly, the results should be interpreted as partial-equilibrium scenario estimates that capture the direct effects of the shock rather than the full macro-financial feedback loop.

Taken together, these results support policy frameworks that are targeted, temporary and grounded in granular information on household balance sheets and consumption patterns. This is in line with IMF (2026) policy recommendations. Our results suggest that the optimal policy mix depends on the objective—household welfare, economic impact, or financial stability—and on country-specific distributions of income, leverage, and savings. Achieving cost-effectiveness of fiscal support policies is crucial in the current environment of high public debt and limited fiscal space in most countries relative to the 2022 energy crisis. Integrating household-level data into macroprudential surveillance and stress-testing can therefore improve both the assessment of systemic risk and the calibration of policy responses to macrofinancial shocks. A natural extension would incorporate additional loan-restructuring options, such as amortization deferral or maturity extension where institutional arrangements permit, as well as endogenous credit supply and general-equilibrium feedback among consumption, employment, firms, and bank lending. These channels are outside the present partial-equilibrium framework, and their net effects are not quantified here. Quantitative magnitudes remain conditional on survey vintage and weighting, end-2025 updating, energy and food pass-through, consumption imputation and behavior, and the representative-portfolio bank mapping.

Table 1. Logit Model to Predict Arrears, Log-Odd Ratios

	(1)		(2)		(3)		(4)		(5)	
DSECTI overburdened = 1	0.512**	(3.11)	0.530**	(3.20)	0.590***	(3.54)	0.764***	(3.46)	0.818***	(3.43)
Saving ratio			0.0000596***	(5.95)	0.0000616***	(5.90)	0.0000712***	(5.71)	0.0000737***	(5.57)
Age					-0.00508	(-0.54)	-0.000550	(-0.03)	-0.00240	(-0.15)
Gender					0.00576	(0.04)	-0.348*	(-2.00)	-0.270	(-1.44)
University					-0.333*	(-2.09)	-0.00629	(-0.03)	0.196	(0.82)
Financial knowledge					0.00376	(0.02)	-0.378+	(-1.65)	-0.433+	(-1.83)
Unemployed					1.002***	(4.37)	1.428***	(4.30)	1.249***	(3.71)
Work experience					0.00504	(0.51)	0.0410**	(2.94)	0.0348**	(2.59)
N. unemployed household members					-0.00863	(-0.10)	0.126	(1.18)	0.00801	(0.07)
N. dependent children					0.0415	(0.58)	0.0710	(0.96)	0.0246	(0.31)
LTV of main residence							0.684*	(2.10)	0.442	(1.04)
Family assistance							1.081**	(3.02)	0.907*	(2.51)
Public assistance							0.372+	(1.77)	0.375+	(1.82)
Leasing							-0.939**	(-3.26)	-0.956**	(-3.11)
Adjustable rate									0.433	(1.57)
Liquid assets									-0.704***	(-3.89)
Credit constraint									1.032***	(4.94)
Inheritance									-0.440*	(-2.44)
Net wealth income ratio									0.0245	(1.08)
Country FE	Yes		Yes		Yes		Yes		Yes	
Income tercile FE	Yes		Yes		Yes		Yes		Yes	
Observations	22788		22595		19344		11895		10488	
Pseudo R-squared	0.067		0.068		0.079		0.098		0.131	
AIC	26672221		26462102		25312448		12437200		10358169	
BIC	26672405		26462294		25312684		12437451		10358438	

Source: 2021 HFCS and authors' calculations. Note: t statistics are shown in parentheses; + p<0.1 * p<0.05 ** p<0.01 *** p<0.001. DSECTI is a binary variable that takes the value of 1 when it exceeds 70 percent and 0 otherwise.

Table 2. Logit Model to Predict Arrears, Predicted Probabilities

	(1)		(2)		(3)		(4)		(5)	
DSECTI overburdened = 0	0.0678	(17.64)	0.0674	(17.51)	0.0654	(16.98)	0.0505	(12.05)	0.0442	(10.87)
DSECTI overburdened = 1	0.108	(6.91)	0.109	(6.89)	0.112	(6.89)	0.103	(4.94)	0.0948	(4.52)
AT	0.0708	(4.82)	0.0703	(4.78)	0.0653	(4.64)	0.0355	(2.50)	0.0363	(2.69)
BE	0.0347	(4.57)	0.0347	(4.55)	0.0363	(4.55)	0.0243	(3.23)	0.0256	(3.21)
CY	0.261	(7.41)	0.244	(6.69)	0.237	(6.63)	0.233	(5.51)	0.201	(4.79)
DE	0.103	(6.59)	0.102	(6.55)	0.104	(6.59)	0.0879	(4.76)	0.0924	(4.67)
EE	0.0949	(8.19)	0.0948	(8.18)	0.0905	(7.77)	0.0675	(4.78)	0.0424	(3.41)
ES	0.0769	(10.75)	0.0770	(10.74)	0.0716	(9.90)	0.0531	(5.86)	0.0398	(5.35)
FR	0.0421	(9.30)	0.0418	(9.24)	0.0393	(8.94)	0.0258	(5.11)	0.0277	(4.42)
GR	0.286	(10.07)	0.287	(10.07)	0.288	(9.69)	0.202	(6.00)	0.191	(5.75)
HR	0.0955	(3.12)	0.0997	(3.11)	0.0950	(3.06)	0.202	(2.28)	0.144	(2.04)
HU	0.215	(12.58)	0.216	(12.58)	0.215	(10.94)	0.131	(6.43)	0.133	(6.13)
IE	0.0703	(9.19)	0.0698	(8.99)						
IT	0.0837	(6.75)	0.0829	(6.67)	0.0799	(6.54)	0.0941	(5.88)		
LT	0.190	(4.42)	0.191	(4.42)	0.227	(4.47)	0.228	(3.82)	0.145	(2.94)
LU	0.0492	(4.97)	0.0506	(4.98)	0.0468	(4.70)	0.0359	(3.48)		
LV	0.0944	(1.96)	0.0943	(1.95)	0.0879	(1.91)	0.0830	(1.72)	0.0599	(1.66)
MT	0.0318	(1.74)	0.0195	(1.43)						
NL	0.0512	(7.52)	0.0514	(7.52)	0.0513	(6.85)	0.0372	(5.68)	0.0333	(4.81)
PT	0.0655	(6.23)	0.0658	(6.23)	0.0613	(6.05)	0.0472	(5.04)	0.0359	(4.13)
SI	0.0245	(2.93)	0.0250	(2.94)	0.0251	(2.93)	0.0292	(1.97)	0.0219	(1.85)
SK	0.0682	(5.00)	0.0641	(4.77)	0.0615	(4.60)	0.0637	(3.58)	0.0644	(3.50)
1st income tercile	0.142	(9.07)	0.140	(8.95)	0.123	(7.30)	0.0818	(5.41)	0.0579	(4.89)
2nd income tercile	0.0792	(11.14)	0.0789	(11.09)	0.0732	(10.22)	0.0520	(7.54)	0.0477	(6.96)
3rd income tercile	0.0483	(9.76)	0.0482	(9.71)	0.0515	(8.96)	0.0481	(7.49)	0.0430	(6.63)
Observations	22788		22595		19344		11895		10488	

Source: 2021 HFCS and authors' calculations. Note: t statistics are shown in parentheses. DSECTI is a binary variable that takes the value of 1 when it exceeds 70 percent and 0 otherwise.

Table 3. Average Country Shocks Across Scenarios (2026)

	Interest rate (percent)	Income growth (percent)	Food price (Index 100=2016)	Energy price (Index 100=2016)	Headline inflation (percent)	Core inflation (percent)	Unempl. (percent)
2025							
Initial conditions	3.06	3.05	127.23	165.40	2.75	3.05	5.94
2026							
Baseline	3.07	2.52	130.18	159.23	2.30	2.52	5.87
Tightening	5.07	2.52	130.18	159.23	2.30	2.52	5.87
Income	3.07	-7.74	130.18	159.23	2.30	2.52	10.87
FSAP geo	5.42	2.88	143.59	192.50	4.84	5.89	6.67
Cost of living	5.07	2.52	156.21	191.07	2.30	2.52	5.87
Middle East war	5.07	2.52	158.84	326.27	3.30	3.52	5.87
Combined	5.07	-7.74	130.18	159.23	2.30	2.52	10.87

Sources: Eurostat, OECD, IMF and authors' calculations. Note: The columns report the following variables: *Interest Rate*, defined as the lending rate applied to new loan originations and the repricing of outstanding loans; *Income Growth*, measured as nominal wage growth (OECD data for 2021–24 and IMF WEO projections for 2025–28); *Food Price*, based on food commodity prices in European markets (IMF Global Assumptions); *Energy Price*, based on fuel and energy commodity prices (IMF Global Assumptions); *Headline Inflation*; *Core Inflation* and *Unemployment* Rate. The rows correspond to the following scenarios: (i) *Baseline*, based on the October 2025 WEO baseline projections; (ii) *Tightening*, which assumes a 200 basis point increase in lending rates relative to the baseline; (iii) *Income*, which assumes a 10 percent decline in nominal gross household disposable income and a 5 percentage point increase in the unemployment rate relative to the baseline; (iv) *FSAP Geo*, corresponding to the geopolitical scenario developed for the 2025 Euro Area FSAP; (v) *Cost of Living*, which combines the *Tightening* scenario with a 20 percent increase in food and energy prices relative to the baseline; (vi) *Middle East War*, corresponding to the severe scenario in the April 2026 WEO; and (vii) *Combined*, which incorporates both the *Tightening* and *Income* scenarios.

References

Ampudia, M., van Vlokhoven, H. and Żochowski, D., (2016), "Financial fragility of euro area households", Journal of Financial Stability, Vol. 27, pp. 250-262.

Arnould, G., Leika, M., Panagiotopoulos, A., Spector, M., and Valderrama, L. (2025), "Stress Testing the Banking Sector", Euro Area FSAP, IMF Country Report No. 25/211 available at EA FSAP TN Stress Testing the Banking Sector

Bandoni, E., Fourné, F., and Jarmulska, B. (2025), "Mortgage loan rates and the defaults of variable rate mortgages", ECB Working Paper No. 3112.

Bank of England (2023), "How are the rising cost of living and interest rates affecting households' ability to pay their mortgage", Bank Overground, January.

Bech, S., Mikkelsen, J. and Hviid, S. (2021), "Measuring household interest-rate sensitivity in Denmark", Danmarks Nationalbank; Working Paper, No. 183, November.

Bruegel (2026), 2026 European energy crisis fiscal response tracker.

Campbell, J. Y. (2006), "Household Finance", *Journal of Finance*, August.

Cocco, J., (2005), "Portfolio Choice in the Presence of Housing", *Review of Financial Studies*, Vol. 18, Issue 2, pp. 535-567.

Crawley, E., and Theloudis, A. (2024), "Income Shocks and Their Transmission into Consumption", Finance and Economics Discussion Series 2024-038. Washington: Board of Governors of the Federal Reserve System, https://doi.org/10.17016/FEDS.2024.038

Dausa i Noguera, N., Dimou, M., and Kouvavas, O. (2026), "Tracing the ripple effects of the Middle East war on euro area consumption", ECB Economic Bulletin, Issue 5/2026.

Dieckelmann, D., and Metzler, J. (2022), "Household inequality and financial stability risks: exploring the impact of changes in consumer prices and interest rates.", ECB, Financial Stability Review, Special Feature, November.

Dirma, M. and Karmelavičius, J. (2025), "Micro-assessment of macroprudential borrower-based measures", Journal of Banking & Finance, 176, 107455.

Druedahl, J., and Jørgensen, T.H., (2020), "Can consumers distinguish persistent from transitory income shocks?", The Economic Journal, 130, 2410-2437, November.

European Central Bank (2022), "Household inequality and financial stability risks: exploring the impact of changes in consumer prices and interest rates", Financial Stability Review, November.

Eurostat (2025), "Housing cost overburden rate by household type – EU-SILC survey", available at Statistics | Eurostat

Foulis, A., Hazell, J., Mian, A., and Tracey, B., (2026), “How do interest rates affect consumption? Household debt and the role of asset prices”, Bank of England, Staff Working Paper No. 1,173.

Gomes, F., Haliassos, M., and Ramadorai, T., (2021), “Household Finance”, Journal of Economic Literature, Vol. 59, No.3, September.

Gross, M., Tressel, T., Ding, X., and Tereanu, E. (2022), “What drives mortgage default risk in Europe and the US? IMF Working Paper 22/65, April.

Gross, M., and Población, J. 2017. “Assessing the Efficacy of Borrower-Based Macroprudential Policy Using an Integrated Micro-Macro Model for European Households,” Economic Modelling, 61:510–28.

Holló, D., and Papp, M. (2007). “Assessing household credit risk: evidence from a household survey”, Magyar Nemzeti Bank Occasional Papers, No. 70.

International Monetary Fund (2026a), “IMF Global Policy Tracker”, IMF Global Policy Tracker: How Countries are Responding to the Economic Impact of the War in the Middle East.

International Monetary Fund (2026b), “Living in a shock-prone world: lessons from cost-of-living crises”, World Economic Outlook, Chapter 2, October.

International Monetary Fund (2026c), “Locked Out: Drivers and Economic Implications of Declining Housing Affordability in Europe”, European Department, forthcoming.
Jurča, P., Klacso, J., Tereanu, E., Forletta, M., Gross, M. (2020), “The effectiveness of borrower-based macroprudential measures: A quantitative analysis for Slovakia”, IMF Working Paper 2020/134.

Krimmel, J., Moore K. B., Sabelhaus, J., and Smith, P. (2013), “The Current State of U.S., Household Balance Sheets”, Federal Reserve Bank of St. Louis Review, September/October, 95(5), pp. 337-59.

Martinello, A., Meyer, B., and Nissen, R. (2023), “Homeowners’ budgets and debt servicing capacity projections”, Danmarks Nationalbank; Economic Memo, No. 4, June.

Mian, A., and Sufi, A. (2014), “House of Debt – How They (and You) Caused the Great Recession, and How We Can Prevent It from Happening Again”, The University of Chicago Press.

Michelangeli, V., and Pietrunti, M. (2014), “A microsimulation model to evaluate Italian households’ financial vulnerability”, International Journal of Microsimulations 7 (3), 53-79 (winter).

Neugebauer, K., Oliveira, V., and Ramos, Â, (2021), “Assessing the effectiveness of the Portuguese borrower-based measure in the Covid-19 context”, Working Paper No. 202110, Banco de Portugal.

Nier, E., Popa, R., Shamloo, M. and Voinea, L. (2019). “Debt Service and Default: Calibrating Macroprudential Policy Using Micro Data”, IMF Working Paper 2019/182.

OECD (2021), “Brick by Brick – Building Better Housing Policies”, May.

Semjonovs, A., Ārins, M., and Sinenko, N. (2022), “Borrowers in the midst of rising costs: what is the capacity of household borrowers to withstand the sharp increase in prices and interest rates?”, Latvijas Banka, Macroeconomic Notes.

Valderrama L, Gorse, P., Marinkov, M., and Topalova, P. (2023), “European Housing Markets at a Turning Point – Risks, Household and Bank Vulnerabilities, and Policy Options”, IMF Working Paper 23/76.

Appendix I. Robustness and Predictive Validation

Comparison with Conventional Vulnerability Indicators

We assess whether the cost-of-living-adjusted debt-service-to-income ratio (DSECTI) provides additional information relative to conventional household vulnerability indicators. Using the common estimation sample and the Model 4 specification, we compare a DSECTI threshold of 70 percent with a DSTI threshold of 40 percent and a DTI threshold of six times annual income. Both DSECTI and DSTI are positively and significantly associated with arrears, whereas DTI is not statistically significant (see Table A1). We find that the estimated marginal effect of DSECTI is larger: the predicted probability of arrears rises from 4.8 to 11.6 percent for households above the DSECTI threshold, compared with an increase from 4.9 to 10.3 percent for the DSTI threshold.

Table A1. Comparison Across Vulnerability Indicators, Odds Ratios

	(1)		(2)		(3)	
	DSECTI>=70%		DSTI>=40%		DTI>=6	
Overburdened = 1	0.963***	(4.20)	0.796**	(3.21)	0.0649	(0.19)
Saving ratio	0.0000683***	(5.45)	0.0000684***	(5.46)	0.0000681***	(5.42)
Age	-0.00499	(-0.29)	-0.00375	(-0.22)	-0.00378	(-0.22)
Gender	-0.325+	(-1.71)	-0.333+	(-1.75)	-0.335+	(-1.77)
University	0.0640	(0.28)	0.0567	(0.25)	0.0671	(0.29)
Financial knowledge	-0.434+	(-1.89)	-0.406+	(-1.78)	-0.406+	(-1.78)
Unemployed	1.391***	(4.23)	1.361***	(4.08)	1.384***	(4.17)
Work experience	0.0391**	(2.71)	0.0383**	(2.66)	0.0389**	(2.69)
N. unemployed household members	-0.00702	(-0.06)	-0.00245	(-0.02)	-0.00769	(-0.07)
N. dependent children	0.0137	(0.17)	0.00584	(0.07)	-0.00323	(-0.04)
Country FE	Yes		Yes		Yes	
Income tercile FE	Yes		Yes		Yes	
Observations	10487		10487		10487	
Pseudo R-squared	0.100		0.095		0.091	
AIC	10731073.0		10783548.3		10831151.3	
BIC	10731305.3		10783780.5		10831383.6	

Source: 2021 HFCS and authors' calculations. Note: t statistics are shown in parentheses; + p<0.1 * p<0.05 ** p<0.01 *** p<0.001.

Out-of-Sample Performance

We further compare the predictive performance of the three indicators using five-fold cross-validation, employing identical folds across specifications. Table A2 shows how DSECTI achieves a mean out-of-sample AUC of 0.710 and a mean Brier score of 0.0604. These results are very close to those obtained for DSTI (AUC of 0.712 and Brier score of 0.0600), while DTI performs slightly less well (AUC of 0.707). Thus, DSECTI and DSTI display comparable out-of-sample discriminatory power, but DSECTI is associated with a larger estimated increase in the probability of arrears. We therefore interpret DSECTI as a complementary measure that incorporates households' residual resources after essential consumption, rather than claiming uniform superiority over all conventional indicators.

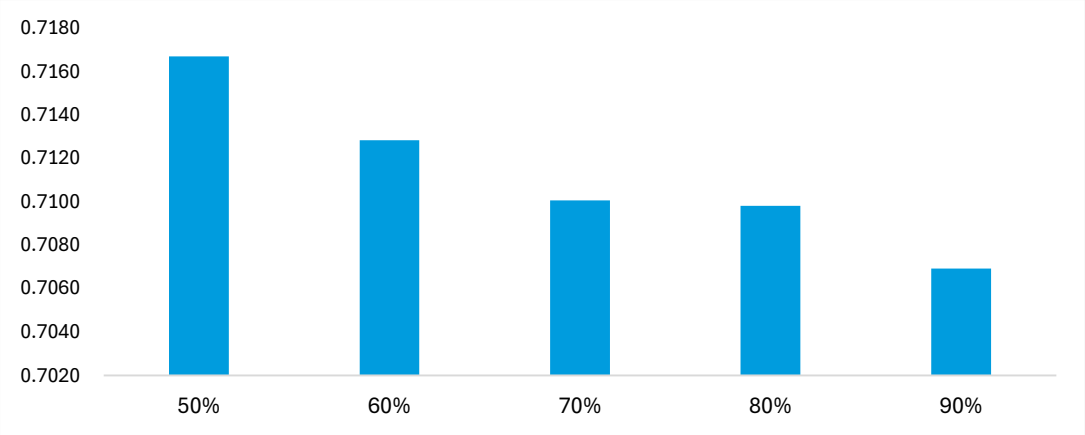
Table A2. Comparison Five-Fold Cross-Validation Performance of Alternative Vulnerability Indicators

Indicator	Mean AUC 5-fold	Mean Brier 5-fold
DSECTI	.7100698	0.06036
DSTI	.7123471	0.05995
DTI	.7069093	0.06018

Sensitivity to the DSECTI Threshold

We validate the DSECTI cut-off out of sample by estimating the Model 4 specification at thresholds ranging from 50 to 90 percent (Figure A1). Mean five-fold cross-validation AUC is highest at the 50 percent threshold (0.717), followed by 60 percent (0.713), 70 percent (0.710), 80 percent (0.710), and 90 percent (0.707). The differences are modest, indicating that predictive discrimination is broadly stable across plausible thresholds. The 70 percent threshold is therefore not selected because it maximizes out-of-sample AUC; rather, it provides the best economically meaningful definition of severe financial strain and is associated with a substantial increase in predicted arrears.

Figure A1. Out-of-sample validation across DSECTI thresholds
Mean AUC 5-Fold



Sources: 2021 HFCS, and authors' calculations. Note: The chart shows the probability of arrears for overburdened households (defined as households with a DSECTI exceeding the specified threshold) relative to non-overburdened households, across a range of DSECTI thresholds. Probabilities are estimated from panel logit regressions controlling for income tercile and country fixed effects.

Common-Sample Estimates

To ensure that differences across model specifications do not reflect changes in sample composition, we re-estimate all baseline models on a common sample of 10,487 observations, defined by non-missing information on the full set of controls and all vulnerability indicators (Table A3). The coefficient on DSECTI overburden remains positive and statistically significant throughout, ranging from 0.981 in the baseline specification to 0.818 in the most comprehensive specification. The corresponding marginal effects remain economically meaningful, with predicted arrears rates for overburdened households consistently exceeding those for non-overburdened

households. These results confirm that the main findings are not driven by the progressive loss of observations as additional controls are introduced.

Table A3. Common-Sample Robustness Estimates, Odds Ratios

	(1)		(2)		(3)		(4)		(5)	
DSECTI overburdened = 1	0.981***	(4.33)	0.979***	(4.32)	0.986***	(4.31)	0.963***	(4.20)	0.818***	(3.43)
Saving ratio			0.0000696***	(5.59)	0.0000655***	(5.24)	0.0000683***	(5.45)	0.0000737***	(5.57)
Age					-0.0108	(-0.70)	-0.00499	(-0.29)	-0.00240	(-0.15)
Gender					-0.354+	(-1.87)	-0.325+	(-1.71)	-0.270	(-1.44)
University					0.0565	(0.24)	0.0640	(0.28)	0.196	(0.82)
Financial knowledge					-0.432+	(-1.89)	-0.434+	(-1.89)	-0.433+	(-1.83)
Unemployed					1.468***	(4.58)	1.391***	(4.23)	1.249***	(3.71)
Work experience					0.0334*	(2.34)	0.0391**	(2.71)	0.0348**	(2.59)
N. unemployed household members					-0.0177	(-0.14)	-0.00702	(-0.06)	0.00801	(0.07)
N. dependent children					0.101	(1.32)	0.0137	(0.17)	0.0246	(0.31)
LTV of main residence							0.562	(1.57)	0.442	(1.04)
Family assistance							0.970*	(2.56)	0.907*	(2.51)
Public assistance							0.449*	(2.08)	0.375+	(1.82)
Leasing							-0.977**	(-3.25)	-0.956**	(-3.11)
Adjustable rate									0.433	(1.57)
Liquid assets									-0.704***	(-3.89)
Credit constraint									1.032***	(4.94)
Inheritance									-0.440*	(-2.44)
Net wealth income ratio									0.0245	(1.08)
Country FE	Yes		Yes		Yes		Yes		Yes	
Income tercile FE	Yes		Yes		Yes		Yes		Yes	
Observations	10487		10487		10487		10487		10487	
Pseudo R-squared	0.062		0.066		0.084		0.100		0.131	
AIC	11188329.9		11138023.9		10925300.7		10731073.0		10358139.8	
BIC	11188467.8		11138169.0		10925503.9		10731305.3		10358408.4	

Source: 2021 HFCS and authors' calculations. Note: t statistics are shown in parentheses; + p<0.1 * p<0.05 ** p<0.01 *** p<0.001. DSECTI is a binary variable that takes the value of 1 when it exceeds 70 percent and 0 otherwise.

Survey Weights and Model Fit

All baseline specifications use HFCS household population weights. Since weighted logit estimation relies on a pseudo-likelihood, AIC and BIC values are affected by the scale of the weights and should not be interpreted as directly comparable across models estimated on different samples or under different weighting schemes. As a cross-check, we re-estimate Model 4 without survey weights. The DSECTI coefficient remains positive and statistically significant, although its magnitude is smaller in the unweighted specification (Table A4). This result confirms the qualitative robustness of the main association, while indicating that population weighting affects its estimated magnitude. Due to nature of the available HFCS data, such a limitation is unavoidable in our proposed methodology.

Rare Events and Regularized Prediction

Finally, we examine whether sparse arrears events affect the main result. A rare-events logit correction yields an essentially unchanged DSECTI coefficient relative to the corresponding standard logit specification (0.782 in both models), indicating that the estimated association is not driven by rare-events bias (Table A5). We also compare the baseline Model 4 logit with a cross-validated LASSO logit on a held-out test sample. The LASSO model modestly improves predictive performance, increasing the AUC from 0.717 to 0.750 and reducing the Brier score from 0.0640 to 0.0612. The improvement suggests that additional regularization can enhance prediction, while the stability of the baseline results supports the use of the more transparent Model 4 specification for the main analysis (Table A6).

Table A4. Weighted and Unweighted Model 4 estimates, Odds Ratios

	(1) Model 4 (weighted)		(2) Model 4 (unweighted)	
DSECTI overburdened = 1	0.963***	(4.20)	0.409**	(3.04)
Saving ratio	-0.00499	(-0.29)	0.0203**	(2.88)
Age	0.0000683***	(5.45)	0.0000632***	(5.58)
Gender	-0.325+	(-1.71)	-0.270**	(-3.06)
University	0.0640	(0.28)	-0.453***	(-4.76)
Financial knowledge	-0.434+	(-1.89)	-0.149	(-1.01)
Unemployed	1.391***	(4.23)	0.868***	(3.88)
Work experience	0.0391**	(2.71)	0.00621	(0.92)
N. unemployed household members	-0.00702	(-0.06)	0.120*	(2.01)
N. dependent children	0.0137	(0.17)	-0.0243	(-0.51)
LTV of main residence	0.562	(1.57)	1.252***	(7.78)
Family assistance	0.970*	(2.56)	0.847***	(4.89)
Public assistance	0.449*	(2.08)	0.321**	(3.03)
Leasing	-0.977**	(-3.25)	-0.113	(-0.68)
Observations	10487		10487	
Pseudo R-squared	0.100		0.128	
AIC	10731073.0		4617.0	
BIC	10731305.3		4849.3	

Source: 2021 HFCS and authors' calculations. Note: t statistics are shown in parentheses; + p<0.1 * p<0.05 ** p<0.01 *** p<0.001. DSECTI is a binary variable that takes the value of 1 when it exceeds 70 percent and 0 otherwise.

Table A5. Rare-events and Regularized-Logit Robustness Checks Odds Ratios

	(1) Standard logit		(2) Relogit (rare events)	
DSECTI overburdened = 1	0.782***	(3.69)	0.782***	(3.70)
Saving ratio	0.0000687***	(5.22)	0.0000687***	(5.24)
Age	-0.00317	(-0.18)	-0.00317	(-0.18)
Gender	-0.303	(-1.62)	-0.303	(-1.62)
University	-0.00368	(-0.02)	-0.00368	(-0.02)
Financial knowledge	-0.158	(-0.80)	-0.158	(-0.80)
Unemployed	1.275***	(3.98)	1.275***	(3.98)
Work experience	0.0389*	(2.41)	0.0389*	(2.42)
N. unemployed household members	0.0153	(0.14)	0.0153	(0.14)
N. dependent children	0.0183	(0.24)	0.0183	(0.24)
LTV of main residence	0.311	(0.89)	0.311	(0.90)
Family assistance	1.042**	(2.79)	1.042**	(2.79)
Public assistance	0.392*	(2.09)	0.392*	(2.10)
Leasing	-0.864**	(-3.11)	-0.864**	(-3.11)
Observations	10487		10487	

Source: 2021 HFCS and authors' calculations. Note: t statistics are shown in parentheses; + p<0.1 * p<0.05 ** p<0.01 *** p<0.001. DSECTI is a binary variable that takes the value of 1 when it exceeds 70 percent and 0 otherwise.

Table A6. Validation against LASSO Model

Model	Test AUC	Test Brier
Logit (Model 4)	0.717261	0.06399
LASSO logit (CV)	0.749963	0.06116



PUBLICATIONS

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